

Mont Terri CD-A Niche 4 OGS Model Report
State of Knowledge, OGS Realization, and Measurement Questions

OGS modelling team report

June 21, 2026

Abstract

This report records the CD-A Niche 4 model in two layers. The first layer is the mathematical Thermo–Richards–Mechanics model: physical setting, unknowns, constitutive assumptions, governing equations, boundary data, and initial state. The second layer is the OGS realization of the same model: project-file blocks, mesh fields, XML parameters, output arrays, and observation-facing quantities. This separation is deliberate. It lets us discuss the model as a scientific object before we decide which material fields, boundary curves, and measurement operators are safe to release for inversion or validation. We use the report as a shared team state-of-knowledge document and as an expert-facing map of the questions that still require provider, measurement, or modelling decisions.

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Chapter 1

Mathematical TRM Model and OGS Realization

1.1 Purpose and Evidence Boundary

We use a two-layer description of the CD-A Niche 4 reference model. The first layer is the mathematical Thermo–Richards–Mechanics (TRM) model: domain, unknowns, constitutive laws, boundary data, and initial state. The second layer is the OGS realization of that model through project-file blocks, XML parameters, mesh fields, curves, time stepping, and VTU outputs. This order is deliberate. It fixes the scientific model before we discuss how each field is passed into OGS or read from OGS results.

The file-level evidence for the implementation is the transferred OGS project `cd_a_open_niche_quad.prj`. Its included process, process-variable, parameter, media, time-loop, solver, and curve XML files are model inputs, not literature sources; when a statement is model-specific, the relevant XML tag or path is stated in the text or tables. For the equations we use the OGS TRM formulation without vapor diffusion, the coupled THM notation of Wang et al. [24, Sec. 2.1–2.3, Eqs. 1, 7–9], and the OGS documentation [19, “Governing equations without vapor diffusion”]. If the OGS project files contain an inactive parameter, commented block, or implementation mismatch, we record it as a gate. We do not promote such entries into the active physics.

1.2 Physical Setting and Modelled Phenomena

The CD-A experiment studies the coupled response of Opalinus Clay around twin niches in the Mont Terri Rock Laboratory. The relevant rock is a bedded claystone with strong hydro-mechanical coupling, structural controls, and a near-niche excavation damaged zone (EDZ). The CD-A work focuses on humidity-driven water-content, pressure, and deformation evolution in the formation [27, abstract] [29, abstract, Sec. 2, Table 1] [30, abstract and Secs. 1–2]. Broader Mont Terri studies show that Opalinus Clay stiffness, strength, pore-pressure response, and damage-zone development depend on suction, bedding, in-situ stress, and fault or fracture structure [1, abstract, Secs. 2–3.3]. EDZ studies further show that hydraulic conductivity and fracture aperture can differ by orders of magnitude from the intact claystone and can evolve through swelling, closure, and self-sealing [11, abstract and Sec. 3.1.1] [14, abstract, Secs. 2.1–2.5].

The reference OGS model represents a controlled subset of this physics. It includes Richards pressure diffusion, capillary retention, relative permeability, porosity storage, orthotropic linear elasticity, Biot/Bishop pore-stress coupling, saturation-dependent swelling stress, and heat-balance terms. It does not explicitly represent vapor diffusion, discrete fracture flow, chemical alteration, sealing/healing kinetics, staged excavation, or full three-dimensional fault geometry. These omissions matter near the niche and in the EDZ, but they remain external gates or later scenario fields in this chapter.

1.3 Mathematical Model

1.3.1 Domain, Unknowns, and State Fields

Let $\Omega \subset \mathbb{R}^2$ be the current two-dimensional CD-A Niche 4 slice, and let $I = [0, t_{\text{end}}]$ be the simulated time interval. The primary unknowns are displacement $\mathbf{u} : \Omega \times I \rightarrow \mathbb{R}^2$, liquid pressure $p_l : \Omega \times I \rightarrow \mathbb{R}$, and

temperature $T : \Omega \times I \rightarrow \mathbb{R}$. Liquid saturation, Darcy velocity, stress, strain, swelling stress, and water-content proxies are state-dependent or post-processed quantities. Tables 1.1–1.3 separate the notation into unknown and output fields, material input fields, and intermediate or derived quantities before they enter the equations.

Table 1.1: Unknowns and OGS output fields used in the reference model.

Field	Symbol	Unit	Mathematical meaning and OGS status
Displacement	$\mathbf{u} = (u_x, u_y)$	m	Primary mechanical unknown and current OGS output field. It describes deformation of the continuum slice relative to the chosen numerical initial state.
Liquid pressure	p_l	Pa	Primary hydraulic unknown and current OGS output field. In the Richards convention used here, negative liquid pressure means positive suction relative to the gas/reference pressure.
Temperature	T	K	Primary thermal unknown in the full TRM process and current OGS output field. In the active parametrization it is constrained to $T_0 = 298.15$ K throughout the domain and time interval.
Liquid saturation	S_l	–	Secondary OGS state/output field computed from pressure through the retention law. It is the mobile OGS liquid occupancy of the pore volume, not total hydrogen-bearing water in the clay.

Table 1.2: Material fields and constitutive inputs to the active model.

Input field or parameter	Symbol	Unit	Material meaning and OGS role
Intrinsic permeability	$\mathbf{K}(\mathbf{x})$	m^2	Material tensor controlling Darcy mobility before multiplication by relative permeability and division by viscosity. The current workflow supplies it through the current OGS mesh field <code>k_i_rd</code> .
Porosity	$n(\mathbf{x})$	–	Continuum pore-volume fraction used in storage, thermal mixing, and the model-side water-content proxy. It is supplied through <code>n_rd</code> and is fixed at 0.105 in the current field package.
Elasticity tensor	$\mathbb{C}$	Pa	Orthotropic linear-elastic stiffness used in the mechanical constitutive law. It is a fixed constitutive input, not a released field in the present workflow.
Biot coefficient and Bishop law	$\alpha_B, b(S_l)$	–	Fixed pore-pressure coupling inputs. In the active XML, $\alpha_B = 1$ and $b(S_l) = S_l$.
Liquid properties	$\rho_l(T, p_l), \rho_l^0(p_l), \mu_l$	$\text{kg m}^{-3}, \text{Pa s}$	Liquid density and viscosity used in storage and Darcy mobility. Under the active isothermal constraint, density depends on pressure but not on temperature.

Table 1.3: Intermediate and derived quantities used by equations and observation operators.

Quantity	Symbol	Unit	Mathematical meaning
Capillary pressure	$p_c = -p_l$	Pa	Retention-curve argument under the active sign convention.

Quantity	Symbol	Unit	Mathematical meaning
Effective saturation	$S_e(p_c)$	–	Normalized saturation variable used by the van Genuchten retention and relative-permeability laws.
Relative permeability	$k_{\text{rel}}(S_l)$	–	Saturation-dependent mobility reduction in the liquid phase. It is computed from the fixed relative-permeability law, not fitted independently.
Darcy velocity and mass flux	$\mathbf{v}_l, \mathbf{q}_l = \rho_l \mathbf{v}_l$	$\text{m s}^{-1}, \text{kg m}^{-2} \text{s}^{-1}$	Derived hydraulic transport quantities. They are not pore velocity unless divided by an explicitly chosen pore-volume factor.
Strain and stress	$\boldsymbol{\varepsilon}(\mathbf{u}), \boldsymbol{\varepsilon}^{\text{th}}(T), \boldsymbol{\sigma}, \boldsymbol{\sigma}^{\text{sw}}$	–, Pa	Mechanical fields derived from displacement, orthotropic elasticity, thermal strain, saturation-dependent swelling stress, and Bishop/Biot pore-pressure coupling.
Swelling stress contribution	$\boldsymbol{\sigma}^{\text{sw}}, p_{\text{sw}}$	–, Pa	Saturation-dependent stress contribution computed from the active swelling-stress-rate law; p_{sw} is the configured swelling pressure.
Volumetric water-content proxy	$\theta = nS_l$	–	Model-derived mobile-water proxy. It needs an observation operator before it can be compared with NMR, ERT, or Taupe/TDR.

We assume a single liquid phase in a porous solid, Richards-type capillary pressure closure, no explicit gas-pressure unknown, and no active vapor diffusion. The reference body force is zero. Temperature is retained in the full TRM statement because it is an OGS primary variable and appears in density, heat storage, and thermo-mechanical coupling [19, “Governing equations without vapor diffusion” and “Process variables”]. Section 1.3.6 then records the reduced isothermal model implied by the temperature initial and boundary entries in `02_process_variables_TRM.xml` and `03_parameters_TRM.xml`.

1.3.2 Liquid Flow, Retention, and Relative Permeability

The liquid velocity is the Darcy velocity. Here k_{rel} is relative permeability, μ_l is liquid viscosity, $\mathbf{K}$ is intrinsic permeability, ρ_l is liquid density, and $\mathbf{q}_l$ is liquid mass flux. With the specific body force disabled, the reference equation is

$$\mathbf{v}_l = -\frac{k_{\text{rel}}(S_l)}{\mu_l} \mathbf{K} \nabla p_l, \quad \mathbf{q}_l = \rho_l \mathbf{v}_l. \quad (1.1)$$

The permeability tensor $\mathbf{K}$ is intrinsic permeability in m^2 . It is not hydraulic conductivity. Hydraulic conductivity is a derived quantity after choosing liquid density, viscosity, relative permeability, and saturation state. The factor nS_l belongs to storage, thermal mixing, and the water-content proxy; it is not part of the Darcy velocity in (1.1).

The active pressure convention is

$$p_c := -p_l. \quad (1.2)$$

Negative liquid pressure therefore means positive suction/capillary pressure relative to the gas/reference pressure used by the Richards formulation. For $p_c > 0$, the configured van Genuchten retention law is written with effective saturation S_e . The parameters S_{lr} , S_{gr} , m , and p_b are the residual liquid saturation, residual gas saturation, retention exponent, and pressure scale listed in Table 1.4:

$$S_e(p_c) = \left[1 + \left(\frac{p_c}{p_b} \right)^{1/(1-m)} \right]^{-m}, \quad S_l = S_{lr} + (1 - S_{lr} - S_{gr})S_e, \quad (1.3)$$

and the corresponding Mualem–van Genuchten relative-permeability law uses the lower cutoff k_{min} :

$$k_{\text{rel}}(S_l) = \max \left(k_{\text{min}}, S_e^{1/2} \left[1 - \left(1 - S_e^{1/m} \right)^m \right]^2 \right). \quad (1.4)$$

Here S_e is the normalized retention variable, whereas S_l is the actual liquid saturation used by storage, water-content proxies, Bishop coupling, and the relative-permeability evaluation. S_{lr} shifts the lower liquid

saturation limit, and S_{gr} changes the conversion from S_e to S_l ; with the current $S_{gr} = 0$, it is still part of the formula but does not add a separate gas-residual correction. This closure follows the standard van Genuchten and Mualem forms [23, p. 892, Eqs. 2–3 and p. 893, Eq. 8] [16, pp. 513–515, Eqs. 1–2, 14, and 21]. In the present workflow these retention and relative-permeability parameters are fixed model structure, not varied calibration fields; their values are declared in `04_2_media_twophase.xml`.

Table 1.4: Parameters introduced by the Darcy, retention, and relative-permeability equations.

XML name or law	Symbol	Active value	Role in Equations (1.1)–(1.4)
<code>k_i/k_i_rd</code>	$\mathbf{K}$	Baseline tensor $\begin{bmatrix} 1.0 & 0 \\ 0 & 0.4 \end{bmatrix} 10^{-19} \text{ m}^2$; current OGS mesh field varies magnitude	Intrinsic permeability tensor in Darcy velocity.
<code>eta</code>	μ_l	$1.0 \times 10^{-3} \text{ Pa s}$	Liquid viscosity in Darcy mobility.
VG saturation law	S_{lr}, S_{gr}, m, p_b	0.1, 0, 0.45, 10 MPa	Retention curve $S_l(p_c)$ with $p_c = -p_l$.
VG relative permeability	$k_{\text{rel}}, k_{\text{min}}$	same residual saturations and $m = 0.45$, with $k_{\text{min}} = 10^{-25}$	Unsaturated mobility reduction multiplying $\mathbf{K}/\mu_l$.

1.3.3 Liquid Mass Balance

The no-vapor liquid mass equation can be written as follows. Overdots denote time derivatives. The coefficient β_l is the pressure-derivative slot associated with the liquid-density law, α_B is the Biot coefficient, α_T^s is the solid thermal-expansivity parameter, and Q_H is the hydraulic source slot:

$$\left(S_l \beta_l - n \frac{\partial S_l}{\partial p_c} \right) \rho_l \dot{p}_l - S_l \left(\frac{\partial \rho_l}{\partial T} + \rho_l (\alpha_B - S_l) \alpha_T^s \right) \dot{T} + \nabla \cdot (\rho_l \mathbf{v}_l) + S_l \alpha_B \rho_l \nabla \cdot \dot{\mathbf{u}} = Q_H. \quad (1.5)$$

The equation shows why permeability, retention, porosity, density, temperature, and mechanical volumetric strain cannot be interpreted independently. In particular, changing the boundary pressure curve or the retention law can mimic a change in permeability through the desaturation timing. The active XML does not add a separate hydraulic source term, so $Q_H = 0$ in the checked baseline and is retained only as the standard equation slot. Vapor transport is also absent from this no-vapor mass balance [19, “Governing equations without vapor diffusion”].

The liquid-density law is kept in its full TRM XML form, but the active run later reduces it to the constant-temperature pressure law:

$$\rho_l(T, p_l) = 1095 \left[1 - 5 \times 10^{-4} (T - 298.15) + 3.2 \times 10^{-10} (p_l - 7.5 \times 10^6) \right]. \quad (1.6)$$

Thus pressure storage, saturation storage, thermal density coupling, and mechanical volumetric coupling belong to the full TRM statement. The paper keeps this equation because it is the checked OGS source form; no thermal calibration or temperature-driven evolution is active in the current comparison. Under the current active temperature constraints, the temperature terms reduce as stated in Section 1.3.6.

Table 1.5: Parameters introduced by the liquid mass balance and density law.

XML name or law	Symbol	Active value	Role in Equations (1.5)–(1.6)
<code>phi/n_rd</code>	n	0.105 in the current field package	Porosity in liquid storage, retention derivative, thermal mixing, and $\theta = n S_l$.
liquid density law	$\rho_l(T, p_l)$	$1095 [1 - 5 \times 10^{-4} (T - 298.15) + 3.2 \times 10^{-10} (p_l - 7.5 \times 10^6)]$	Full TRM density law. Under the active isothermal constraint it reduces to $\rho_l^0(p_l)$ in Equation (1.14).

XML name or law	Symbol	Active value	Role
pressure slope in liquid density law	β_l	$3.2 \times 10^{-10} \text{ Pa}^{-1}$ relative pressure slope in the linear density law	Used in the pressure-storage slot of Equation (1.5); not a separately released calibration parameter.
biot	α_B	1	Couples volumetric strain and pore pressure to hydraulic storage and mechanics.
solid thermal expansivity	α_T^s	CTE=1254.74 in XML	Appears in the thermal-coupling term of Equation (1.5); value and unit remain the CTE provenance caveat recorded in Table 1.7.
hydraulic source	Q_H	No separate active source in the process file	Retained as the standard equation slot, but not used as an additional forcing term.

1.3.4 Quasi-static Mechanics

The mechanical balance is quasi-static. The vector $\mathbf{f}$ is the body-force term, $b(S_l)$ is the Bishop saturation factor, and α_B is the Biot coefficient:

$$\nabla \cdot (\boldsymbol{\sigma} - b(S_l)\alpha_B p_l \mathbf{I}) + \mathbf{f} = \mathbf{0}. \quad (1.7)$$

The Bishop stress factor $b(S_l)$ represents the saturation-dependent weighting of the pore-pressure term in unsaturated effective stress [4, bibliographic record, no DOI found]. For the reference setup, the OGS Bishop law has exponent one and the Biot coefficient is one, so $b(S_l) = S_l$ and $\alpha_B = 1$ after substitution [19, TRM momentum equation and media-property table]; the same values are declared by `BishopsPowerLaw` and `biot` entries in `04_2_media_twophase.xml` and `03_parameters_TRM.xml`. We keep the symbolic form in (1.7) because it makes the coupling assumptions explicit.

The effective stress follows orthotropic linear elasticity with the thermal strain term and the active OGS swelling-stress contribution:

$$\boldsymbol{\sigma} = \mathbb{C} : (\boldsymbol{\varepsilon}(\mathbf{u}) - \boldsymbol{\varepsilon}^{\text{th}}(T)) + \boldsymbol{\sigma}^{\text{sw}}(S_l), \quad \boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2} (\nabla \mathbf{u} + \nabla \mathbf{u}^\top). \quad (1.8)$$

We therefore treat $\boldsymbol{\sigma}^{\text{sw}}$, not an independently specified $\boldsymbol{\varepsilon}^{\text{sw}}$, as the active OGS object. In each material direction, the configured saturation-dependent swelling-stress-rate law is represented schematically by

$$\dot{\sigma}_i^{\text{sw}} = -\lambda_i p_i S_e^{\lambda_i-1} \dot{S}_e, \quad \frac{\Delta \sigma_i^{\text{sw}}}{\Delta t} = -\lambda_i p_i S_e^{\lambda_i-1} \frac{\Delta S_e}{\Delta t}. \quad (1.9)$$

The checked XML gives $p_i = 1.0 \text{ MPa}$ in all directions and exponent $\lambda_i = 1$, with the saturation limits recorded in Table 1.6. The 1.0 MPa value is therefore a checked OGS/XML material input, not a literature-calibrated swelling pressure; changing it would be a separate mechanics model-change decision. This is a two-dimensional slice model. The orthotropic stiffness values come from the source OGS setup and are used as fixed in-plane constitutive data for the current slice; out-of-plane stress assumptions, true 3D transfer of elastic parameters, and excavation unloading are separate mechanics gates rather than active calibration variables. The checked OGS input sets the specific body force to zero, so $\mathbf{f} = \mathbf{0}$ in the active baseline. This is not a staged excavation model. The current mechanics model starts from the already open two-dimensional geometry and has no active initial stress tensor or active top load. Any future mechanical residual therefore requires a separate decision about prestress, excavation unloading, and the reference displacement state. This follows directly from `01_processes_TRM.xml`, where `initial_stress` is commented out, and from `02_process_variables_TRM.xml`, where the top-load Neumann block is commented out.

Table 1.6: Parameters introduced by the quasi-static mechanics equations.

XML name or law	Symbol	Active value	Role in Equations (1.7)–(1.8)
specific_body_force	$\mathbf{f}$	(0, 0)	Body force is disabled in the momentum equation.

XML name or law	Symbol	Active value	Role
biot	α_B	1	Pore-pressure coupling in the effective stress term.
BishopsPowerLaw	$b(S_l)$	S_l	Effective-stress scaling because the configured exponent is one.
E	E_i	(13.8, 6.0, 13.8) GPa	Orthotropic Young moduli in $\mathbb{C}$.
G	G_i	(4.79, 2.46, 4.79) GPa	Orthotropic shear moduli in $\mathbb{C}$.
nu	ν_i	(0.44, 0.22, 0.44)	Orthotropic Poisson ratios in $\mathbb{C}$.
swelling stress rate	p_{sw}	1.0 MPa in each direction	Direction-wise swelling-stress-rate law with exponent one and limits $0.1 \leq S_l \leq 1$.
ic_sigma0, load_top	σ_0 , top load	Defined but inactive	Prestress and top traction are provenance entries, not active mechanical inputs.

1.3.5 Heat Balance

Let $(\rho c_p)^{\text{eff}}$ be effective heat storage and $\mathbf{k}_T^{\text{eff}}$ the effective thermal-conductivity tensor. With liquid heat capacity c_l and thermal source slot Q_T , the TRM heat balance is

$$(\rho c_p)^{\text{eff}} \dot{T} - \nabla \cdot (\mathbf{k}_T^{\text{eff}} \nabla T) + \rho_l c_l \nabla T \cdot \mathbf{v}_l = Q_T. \quad (1.10)$$

The effective heat storage and thermal conductivity use porosity and saturation mixing:

$$(\rho c_p)^{\text{eff}} = (1 - n) \rho_s c_s + n S_l \rho_l c_l, \quad \mathbf{k}_T^{\text{eff}} = (1 - n) k_s \mathbf{I} + n S_l k_l \mathbf{I}. \quad (1.11)$$

The active temperature initial and boundary values are both 298.15 K. In the checked XML, the temperature Dirichlet condition is attached to `bulk_all`; `cd_a_open_niche_quad.prj` lists `bulk_all.vtu` as a project mesh. This boundary support is therefore read as the domain-wide temperature constraint used by the active run. Thermal parameters are therefore fixed coupling context rather than fitted thermal calibration fields. The CTE=1254.74 XML value remains a provenance caveat because it is numerically identical to `c_p_s` and carries a heat-capacity-like XML comment in `03_parameters_TRM.xml`.

Table 1.7: Parameters introduced by the heat balance and thermal mixing equations.

XML name or law	Symbol	Active value	Role in Equations (1.10)–(1.11)
rho_s	ρ_s	$2.52 \times 10^3 \text{ kg m}^{-3}$	Solid density in effective heat storage.
c_p_s	c_s	$1254.74 \text{ J kg}^{-1} \text{ K}^{-1}$	Solid heat capacity in effective heat storage.
kappa_th_s	k_s	$2.156 \text{ W m}^{-1} \text{ K}^{-1}$	Solid thermal conductivity in effective mixing.
c_p_l	c_l	$4160 \text{ J kg}^{-1} \text{ K}^{-1}$	Liquid heat capacity in storage and advective heat transport.
kappa_l	k_l	$0.6 \text{ W m}^{-1} \text{ K}^{-1}$	Liquid thermal conductivity in effective mixing.
CTE	α_T^s	1254.74 in XML	Thermal-expansivity parameter used by OGS; value and unit remain a provenance caveat.
thermal source	Q_T	No separate active source in the process file	Retained as the standard equation slot, but not used as an additional forcing term.

1.3.6 Isothermal Reduced Model for the Active Parametrization

The active OGS process is still TRM, and T is still a primary variable in the project file. However, the active initial and boundary data prescribe

$$T(\mathbf{x}, t) = T_0 := 298.15 \text{ K} \quad \text{for all } \mathbf{x} \in \Omega, t \in I. \quad (1.12)$$

Therefore

$$\dot{T} = 0, \quad \nabla T = \mathbf{0}, \quad \varepsilon^{\text{th}}(T) - \varepsilon^{\text{th}}(T_0) = \mathbf{0}. \quad (1.13)$$

Flow and displacement changes do not generate temperature changes in this active model, because the temperature degree of freedom is constrained. The heat equation is solved by the constant field and does not add an independent evolution equation for the present parametrization [19, TRM heat equation and process-variable list].

For interpretation and parameter fitting, the active model can therefore be read as an isothermal Richards–mechanics model with primary unknowns $(\mathbf{u}, p_l)$. The liquid density reduces from (1.6) to

$$\rho_l^0(p_l) = 1095 \left[1 + 3.2 \times 10^{-10} (p_l - 7.5 \times 10^6) \right], \quad (1.14)$$

so density can still vary with pressure, but not with temperature. The liquid mass balance becomes

$$\left(S_l \beta_l - n \frac{\partial S_l}{\partial p_c} \right) \rho_l^0 \dot{p}_l + \nabla \cdot (\rho_l^0 \mathbf{v}_l) + S_l \alpha_B \rho_l^0 \nabla \cdot \dot{\mathbf{u}} = Q_H, \quad (1.15)$$

with the same Darcy velocity (1.1), retention law (1.3), and relative-permeability law (1.4). The mechanical balance keeps the same pore-pressure coupling, while the thermal-strain increment drops out:

$$\nabla \cdot (\boldsymbol{\sigma} - b(S_l) \alpha_B p_l \mathbf{I}) + \mathbf{f} = \mathbf{0}, \quad \boldsymbol{\sigma} = \mathbb{C} : (\boldsymbol{\varepsilon}(\mathbf{u})) + \boldsymbol{\sigma}^{\text{sw}}(S_l). \quad (1.16)$$

This reduced form is not a different OGS project. It is the mathematical model that the current TRM project realizes once the homogeneous constant temperature constraint is applied. Density, viscosity, heat capacities, thermal conductivities, and thermal expansivity are fixed XML inputs in this reading. The active comparison does not fit thermal or liquid-property constants, and the CTE=1254.74 entry is kept only as checked input provenance rather than interpreted as a calibrated thermal-expansion value.

1.3.7 Mathematical Boundary and Initial Data

Let the outer square boundary be

$$\Gamma_{\square} := \Gamma_A \cup \Gamma_B \cup \Gamma_C \cup \Gamma_D,$$

and let Γ_E denote the open-niche boundary. The reference mathematical hydraulic boundary inventory is

$$p_l = p_{\square} = 1.5 \times 10^6 \text{ Pa} \quad \text{on } \Gamma_{\square}, \quad p_l = p_E(t) \quad \text{on } \Gamma_E. \quad (1.17)$$

Thus the reference mathematical boundary inventory uses a Dirichlet hydraulic condition on all four sides of the square. The niche boundary is the only time-dependent hydraulic Dirichlet boundary. Figure 1.1 plots the tabulated $p_E(t)$ data read from `08_08_open_niche_seasonal.xml`. This is the model-time boundary-data vector used by the current project, not the commented shifted calendar-time variant. The negative values of $p_E(t)$ are interpreted through (1.2): they are liquid-pressure values corresponding to positive suction [30, Sec. 3, Eqs. 1–2].

The figure is an input diagnostic, not a measurement-provenance closure. It shows the numerical forcing passed to OGS. The source sensors, filtering, time-origin convention, and uncertainty of this boundary-data vector remain the boundary-input gate recorded in Section 1.5.

The mechanical boundary inventory is component-wise kinematic support on the outer square,

$$u_x = 0 \quad \text{on } \Gamma_A \cup \Gamma_B, \quad u_y = 0 \quad \text{on } \Gamma_C \cup \Gamma_D, \quad (1.18)$$

with natural tangential components and natural niche traction in the current model. On the niche boundary this means no prescribed mechanical displacement and no separately applied mechanical traction. It does not remove hydraulic coupling: the prescribed niche liquid pressure still enters the momentum balance through the Bishop/Biot pore-pressure term in (1.7). The thermal boundary is $T = 298.15 \text{ K}$ on the thermal support used by the OGS setup. The initial conditions are

$$\mathbf{u}(\mathbf{x}, 0) = \mathbf{0}, \quad p_l(\mathbf{x}, 0) = 1.5 \times 10^6 \text{ Pa}, \quad T(\mathbf{x}, 0) = 298.15 \text{ K}. \quad (1.19)$$

The pressure expression in the XML contains a vertical term multiplied by zero. Consequently, the active initial pressure field is spatially uniform and does not encode a drained near-niche post-excavation pressure field.

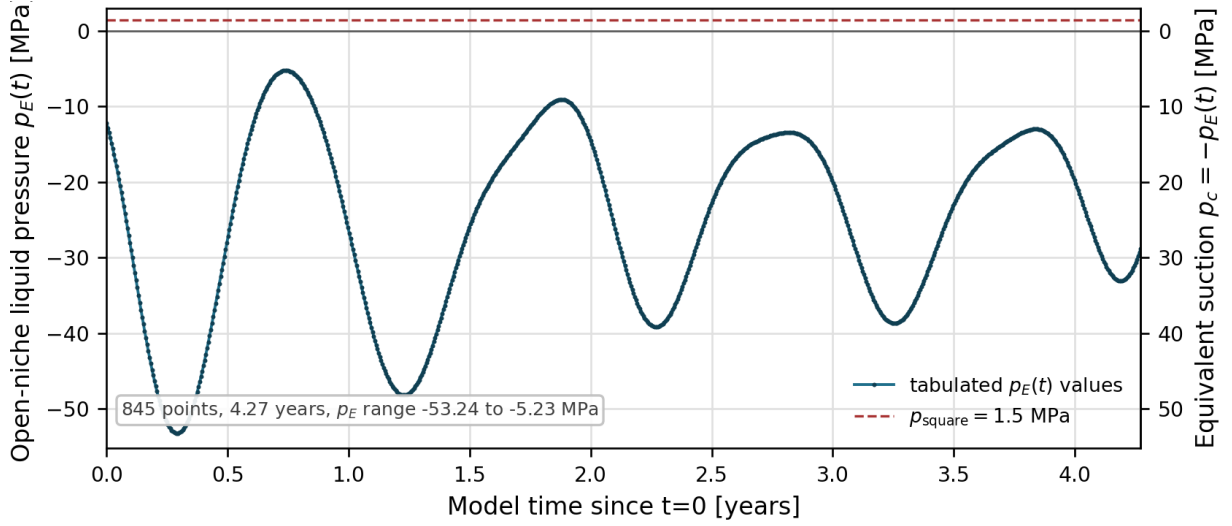


Figure 1.1: Tabulated open-niche pressure boundary data $p_E(t)$ prescribed on Γ_E . The markers are the 845 XML values used by the active OGS setup. The left axis shows liquid pressure; the right axis shows the equivalent suction under $p_c = -p_E(t)$. The dashed line marks the constant square-boundary pressure $p_{\square} = 1.5$ MPa for scale.

1.4 OGS System Realization

1.4.1 Project Blocks and Process Variables

The OGS realization is the `THERMO_RICHARDS_MECHANICS` process in `01_processes_TRM.xml`. The project assigns polynomial order 2 to displacement and order 1 to pressure and temperature. The same process block exposes `velocity`, `saturation`, `sigma`, `epsilon`, and `swelling_stress` as secondary variables. Gravity is off because `specific_body_force` is `0 0` [19, “Process variables”] [18, initial and boundary-condition structure].

The time-loop files bind this process to Backward Euler time discretization, fixed time stepping, and the nonlinear solver `basic_newton`. The nonlinear solver is a Newton solver with at most 150 iterations and delegates linear systems to `general_linear_solver`, which uses `PardisoLU`. The convergence criterion is component-wise Δx in the L^2 norm, with absolute tolerance 10^{-10} and relative tolerance 10^{-8} for each primary-variable component. These settings define the numerical realization of the frozen TRM equations; they do not change the governing equations or the material-field choices.

Table 1.8 maps the mathematical quantities to the OGS blocks, parameters, mesh fields, and output arrays. This table is the bridge between the mathematical model and the file-level system. It also prevents us from treating an observation operator, such as NMR water content or ERT resistivity, as if it were an OGS primary variable.

Table 1.8: Mapping between mathematical quantities and the OGS realization.

Mathematical quantity	OGS block, tag, or field	Input realization	Output or downstream use
$\mathbf{u}$	Process variable <code>displacement</code> in <code>02_process_variables_TRM.xml</code> .	Two components, polynomial order 2, initial parameter <code>ic_displacement</code> ; component-wise Dirichlet supports on the outer square.	Current OGS VTU array <code>displacement</code> . Mechanical observations need a reference-zero and support operator.
p_t	Process variable <code>pressure</code> .	One component, polynomial order 1, initial parameter <code>ic_pressure</code> ; Dirichlet pressure data on the reference hydraulic supports.	Current OGS VTU array <code>pressure</code> . Pressure measurements and RH-derived suction require sign and reference-pressure conventions.

Mathematical quantity	OGS block, tag, or field	Input realization	Output or downstream use
T	Process variable temperature .	One component, polynomial order 1, initial parameter ic_temperature ; Dirichlet temperature parameter bc_temperature_outside .	Current OGS VTU array temperature . In the active parametrization it is constrained to $T_0 = 298.15\text{ K}$, so thermal gradients and temperature-change terms vanish.
K	Medium property permeability , parameter k_i .	Current OGS MeshElement field k_i_rd on the projected bulk mesh. The source package also contains a constant tensor fallback.	Not a dynamic state variable. Observation operators sample or average the mesh field for pulse-test permeability comparisons.
n	Medium property porosity , parameter phi .	Current OGS MeshElement field n_rd ; current field package is spatially constant at 0.105.	Current OGS VTU array porosity . Used with saturation to form $\theta = nS_l$.
S_l	Medium property saturation and process secondary variable.	Computed by SaturationVanGenuchten from $p_c = -p_l$; not prescribed as a primary variable.	Current OGS VTU array saturation . Main state field for water-content observation operators.
k_{rel}	Medium property relative_permeability .	Computed by the OGS van Genuchten relative-permeability law; fixed parameters in the reference setup.	Usually not exported as a required VTU field. It affects pressure and saturation through Darcy mobility.
v_l and q_l	Secondary variable velocity .	Computed from K , k_{rel} , μ_l , and ∇p_l .	Available from the process if requested. Flux residuals require an explicit Darcy-velocity versus pore-velocity convention.
σ , ε , swelling stress	Secondary variables sigma , epsilon , and swelling_stress .	Computed from the orthotropic elasticity law, Bishop/Biot coupling, thermal strain, and saturation-dependent swelling.	Available from the process if requested. Current HM measurements remain gated by prestress, excavation, and reference-zero decisions.
$p_E(t)$	open_niche_seasonal and open_niche_seasonal_curve .	CurveScaled parameter with unit scaling factor 1.0.	Active boundary forcing in the reference setup. Its measurement provenance, filtering, and uncertainty remain open.

1.4.2 Mesh and Boundary Supports

The transferred project uses boundary submeshes **cd-a_left**, **cd-a_right**, **cd-a_top**, **cd-a_bottom**, and **cd-a_niche4**. Figure 1.2 gives the schematic notation for these supports. The figure is only a notation aid; the boundary-condition table below gives the model-facing mapping to XML tags and prescribed data.

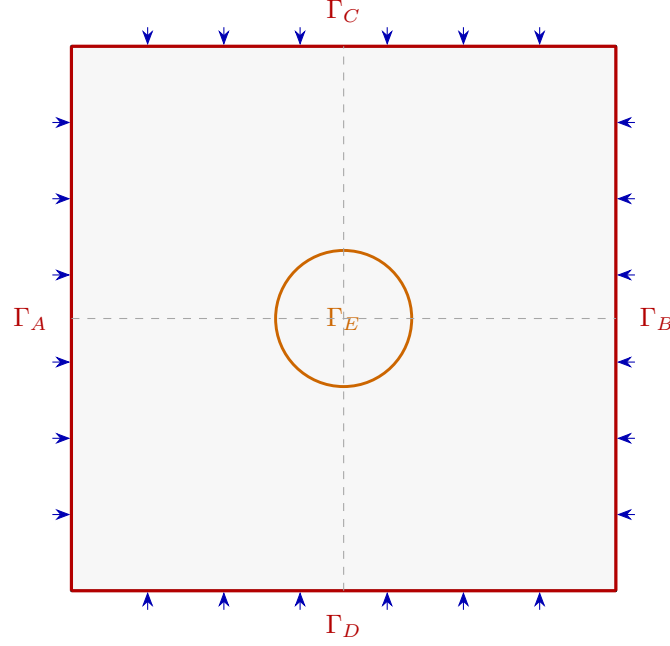


Figure 1.2: Schematic boundary-condition map for the reference two-dimensional model. The outer square boundary is denoted by $\Gamma_A \cup \Gamma_B \cup \Gamma_C \cup \Gamma_D$, and the niche boundary is denoted by Γ_E .

Table 1.9: Boundary-set notation, XML tags, and prescribed mechanical/hydraulic data.

Boundary set	XML tag name and parameter	Mechanical BC	Hydraulic BC
Γ_A	cd-a_left; bc_displacement; reference square-pressure parameter bc_pressure_outside.	Dirichlet $u_x = 0$; tangential component natural.	Dirichlet $p_l = p_\square$, $p_\square = 1.5 \times 10^6$ Pa.
Γ_B	cd-a_right; bc_displacement; reference square-pressure parameter bc_pressure_outside.	Dirichlet $u_x = 0$; tangential component natural.	Dirichlet $p_l = p_\square$, $p_\square = 1.5 \times 10^6$ Pa.
Γ_C	cd-a_top; bc_displacement; bc_top_pressure.	Dirichlet $u_y = 0$; tangential component natural; load_top inactive.	Dirichlet $p_l = p_\square$, $p_\square = 1.5 \times 10^6$ Pa.
Γ_D	cd-a_bottom; bc_displacement; reference square-pressure parameter bc_pressure_outside.	Dirichlet $u_y = 0$; tangential component natural.	Dirichlet $p_l = p_\square$, $p_\square = 1.5 \times 10^6$ Pa.
$\Gamma_A \cup \Gamma_B$	cd-a_left, cd-a_right.	Compact mechanics notation: $u_x = 0$.	Part of the square hydraulic Dirichlet boundary $p_l = p_\square$.
$\Gamma_C \cup \Gamma_D$	cd-a_top, cd-a_bottom.	Compact mechanics notation: $u_y = 0$.	Part of the square hydraulic Dirichlet boundary $p_l = p_\square$.
$\Gamma_\square =$ $\Gamma_A \cup \Gamma_B \cup \Gamma_C \cup \Gamma_D$	cd-a_left, cd-a_right, cd-a_top, cd-a_bottom.	Outer-box kinematic support, component-wise.	Dirichlet $p_l = p_\square = 1.5 \times 10^6$ Pa on all four sides of the outer square in the reference boundary inventory. The seasonal pressure curve is not used on the square boundary.
Γ_E	cd-a_niche4; open_niche_seasonal; open_niche_seasonal_curve; pressure_scaling_factor.	No prescribed displacement or traction in the active boundary-condition block.	Dirichlet $p_l = p_E(t)$. The curve has 845 tabulated values, spans 0 to 1.3482×10^8 s, and ranges from about -5.23 to -53.24 MPa.

The table states the required reference boundary inventory: constant hydraulic Dirichlet data on all four sides of the square and a time-dependent niche pressure curve. In the checked `02_process_variables_TRM.xml`, the named pressure conditions are the constant `bc_top_pressure` entry and the `open_niche_seasonal` entry on `cd-a_niche4`. Thus the live modelling distinction is simple: the open niche uses the seasonal curve, while the square boundary is treated as fixed-pressure support. The remaining open question is not which boundary is seasonal, but whether the seasonal curve’s source timestamps and model-time origin match the measurement time axes used later.

1.4.3 Material Fields and OGS Inputs

The OGS parameter block can define constants, functions, curve-scaled data, and mesh element or mesh node fields [17, sections “CurveScaled”, “Function”, and “MeshElement and MeshNode”]. The current OGS

workflow uses **MeshElement** parameters for intrinsic permeability and porosity. The intrinsic permeability tensor is supplied through **k_i_rd**; this is the local workflow field whose magnitude is varied in the present inversion examples. Candidate changes enter OGS through this current-run mesh field or through explicit XML/input replacement, not through hidden edits to the governing equations. Porosity is supplied through **n_rd**, but in the current field package all cells have value 0.105. This is a mesh-field representation of fixed support, not a porosity inversion. A heterogeneous porosity field would be a new modelling decision, not a property already varied by the current **n_rd** package. Likewise, $\alpha_B = 1$ and the Bishop exponent one are checked XML choices; the paper does not claim that this is the only valid HM setup. The current local workflow varies only the permeability tensor field across candidate inverse runs; other families remain present in OGS but are not fitted in this pass. If the permeability tensor components encode bedding anisotropy through a rotation, the bedding angle and coordinate convention must be stated explicitly. The present text therefore treats bedding orientation as a structural prior and a tensor-parameterization gate, not as a hidden fact implied by component values.

Table 1.10: Material and constitutive fields in the active model or tied to OGS fields.

Field	Material meaning	Relation to other fields	OGS role in the current workflow
Intrinsic permeability tensor $\mathbf{K}(\mathbf{x})$	Clay/EDZ effective Darcy mobility before relative permeability and viscosity are applied. It is the main hydraulic material field for the present inversion.	Hydraulic conductivity is derived from $\mathbf{K}$, liquid density, viscosity, relative permeability, and saturation state; it is not an independent base field here.	Current OGS MeshElement field k_i_rd . Current release gates allow permeability magnitude changes while tensor shape remains staged.
Porosity $n(\mathbf{x})$	Effective pore-volume fraction used by the OGS continuum model.	Water content is $\theta = nS_l$ only for the mobile/OGS liquid storage; NMR, ERT, and Taupe/TDR may also see bound/interlayer water or calibration offsets.	Current OGS MeshElement field n_rd , fixed at 0.105 in the current field package. Not released as an unknown yet.
Saturation S_l and retention $S_l(p_c)$	Liquid occupancy of the OGS pore space, governed by a retention law.	$p_c = -p_l$ under the active gas/reference convention; the same law controls storage derivative and water-content output.	Fixed OGS van Genuchten saturation law with $S_{lr} = 0.1$, $S_{gr} = 0$, exponent $m = 0.45$, and $p_b = 10$ MPa.
Relative permeability $k_{rel}(S_l)$	Reduction of liquid mobility in unsaturated conditions.	Multiplies $\mathbf{K}/\mu_l$ in Darcy velocity; should not be fitted independently of retention without a release decision.	Fixed OGS van Genuchten relative-permeability law with the same residual saturations, exponent 0.45, and minimum 10^{-25} .
Biot and Bishop coupling	Poromechanical coupling between pore pressure, saturation, and stress.	$\alpha_B = 1$ and $b(S_l) = S_l$ reduce the pore-stress scaling to $S_l p_l \mathbf{I}$.	Fixed XML parameter biot =1 and BishopsPowerLaw exponent 1.
Orthotropic elasticity $\mathbb{C}$	Directional stiffness of Opalinus Clay in the 2D mechanical approximation.	Controls stress/displacement response together with pore pressure, saturation, swelling, and any prestress assumption.	Fixed orthotropic linear elasticity: $E = (13.8, 6.0, 13.8)$ GPa, $G = (4.79, 2.46, 4.79)$ GPa, $\nu = (0.44, 0.22, 0.44)$. Mechanical release is gated.
Swelling stress contribution	Saturation-driven mechanical source representing swelling response.	Coupled to saturation and constrained by mechanical boundary assumptions; can mimic deformation from pressure changes if released prematurely.	Active fixed saturation-dependent swelling law with 1.0 MPa swelling pressures in all directions and limits $S_l \in [0.1, 1]$.

Field	Material meaning	Relation to other fields	OGS role in the current workflow
Thermal and liquid properties	Solid/liquid storage, conductivity, density, viscosity, and isothermal coupling context.	In the full XML, liquid density depends linearly on T and p_l , and thermal expansion can enter THM coupling. In the active reduction, $T = T_0$, so only the pressure part of density varies.	Fixed XML constants/laws. The CTE=1254.74 value is a provenance caveat and should not be interpreted as a calibrated thermal-expansion coefficient.
Fault, fracture, crack, and EDZ fields	Structural features that may localize permeability, porosity, stiffness, or aperture.	Measured aperture could imply hydraulic aperture or localized permeability only after scale, clay-contact, and 3D-to-2D projection assumptions are accepted.	Not an active OGS geometry/material class in this chapter. They remain structural priors, masks, scenarios, or qualitative diagnostics until source data are resolved.
Time-dependent material fields	Possible healing, sealing, fracture closure, or EDZ evolution over the multi-year experiment.	Could appear as $K(\mathbf{x}, t)$, $n(\mathbf{x}, t)$, retention changes, or boundary/input changes, but may be confounded with forcing and saturation state.	Excluded from the first static permeability-field workflow. Release requires same-support repeated evidence and state-output diagnostics.

The material-field question is therefore not only which constants appear in XML. For each field we need to know its material meaning, whether it is a base field or a derived field, whether it is independent enough to vary, and how OGS consumes it. In the first workflow, $\mathbf{K}(\mathbf{x})$ is the only varied base material field. Hydraulic conductivity, water content, storativity, resistivity, and flux are derived from $\mathbf{K}$, n , S_l , fluid properties, and observation operators [19, Darcy law and effective-storage equation] [30, Sec. 3, Eq. 3]. No separate storativity field is varied in this setup. Storage behaviour is already represented through porosity, the saturation derivative, liquid compressibility/density, and hydro-mechanical volumetric coupling in (1.5) and (1.15).

1.4.4 Initial Conditions, Time Grid, and Outputs

The initial-condition XML entries realize (1.19). The time-step file comments identify $t = 0$ with 18 September 2019. The active grid starts at zero seconds, uses fixed Backward Euler stepping, and ends at 1.4×10^8 s. The first block uses 365 steps of 864,000 s, and the second block uses 365 steps of 1,314,900 s. The output schedule is intended to match the monthly ERT cadence after the first output, as stated in comments in 05_time_loop_TRM.xml and 05_1_fixed_timestepping.xml.

Table 1.11: Initial conditions used by the active TRM process.

Variable	XML parameter	Active value	Interpretation and modelling gate
$\mathbf{u}(\mathbf{x}, 0)$	ic_displacement	(0, 0) m	The model starts from zero displacement on the already open two-dimensional geometry. No excavation/driving phase is represented explicitly.
$p_l(\mathbf{x}, 0)$	ic_pressure	1.5×10^6 Pa everywhere	The XML function contains a vertical term multiplied by zero, therefore the active initial liquid pressure is spatially uniform. This remains a gate for pressure history: no post-excavation drained niche field is encoded.
$T(\mathbf{x}, 0)$	ic_temperature	298.15 K	The model is initialized at T_0 . Together with the bulk_all temperature Dirichlet condition, this enforces homogeneous constant temperature in the active parametrization.

Variable	XML parameter	Active value	Interpretation and modelling gate
σ_0	ic_sigma0	Defined but inactive	The file defines in-situ stress-like expressions, but the process XML comments out the initial-stress reference. Prestress is therefore a provenance question, not an active initial condition in the present run.

This initial state is a numerical post-excavation state. It is not a simulation of the initial excavation or gradual niche unloading. In mechanics, `ic_sigma0` is defined but not connected as active initial stress, and `load_top` remains inside a commented top-load block. In hydraulics, the initial pressure is uniform and does not reconstruct near-niche drainage from nearby holes, RH/suction, or direct pressure measurements. Early transients may therefore mix physical drainage with adjustment from the prescribed initial state. The uniform initial pressure is therefore a checked numerical input, not a claim that the nearby experiments prove spatially uniform post-excavation drainage. The elapsed time between excavation and this $t = 0$ date is not closed by the checked OGS files; it remains part of the excavation-reference gate.

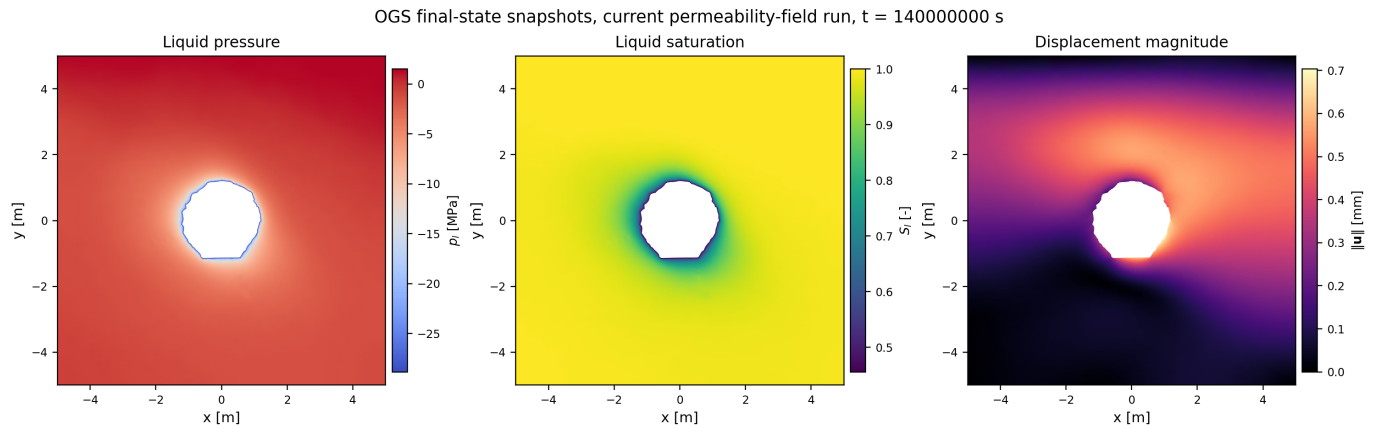


Figure 1.3: Final-state snapshots from the current permeability-field OGS run at $t = 1.4 \times 10^8$ s. The panels show liquid pressure, liquid saturation, and displacement magnitude from the current OGS VTU output. The figure is a diagnostic visualization of the active solver state; it is not itself a measurement residual.

1.4.5 Model Outputs and Observation Operators

The OGS output layer must stay separate from measurement-specific observation operators. The current OGS workflow requests pressure, saturation, temperature, displacement, and porosity. These arrays are model states or fixed support fields, not residuals by themselves. A residual is the signed model-data mismatch, for example $r = H(u) - d$, after the observation operator H has fixed the measured quantity, support, time mapping, unit convention, reference state, and uncertainty. For a Gaussian likelihood this becomes a posterior-informing term only after the uncertainty is stated, e.g. $(H_i(u) - d_i)/\sigma_i$ or $-\frac{1}{2}(H_i(u) - d_i)^2/\sigma_i^2$ for datum i . Permeability comparisons are a separate case: they sample or average the input tensor field, because permeability is not a dynamic state output.

Table 1.12: Model outputs and observation-facing quantities.

Quantity	OGS status	Measurement tie	Required operator or gate
p_l	Primary variable and current OGS VTU output.	Direct pressure or mini-piezometer data, pressure-boundary checks, and pressure diagnostics.	Reference convention, elevation/head conversion, gauge/absolute convention, and 2D projection error must be explicit.
u	Primary variable and current OGS VTU output.	Geoscope, crackometer, levelling, extensometer, laser-scan, and deformation screens.	Reference-zero, sign convention, support geometry, and whether displacement can be used as a hard residual are unresolved.

Quantity	OGS status	Measurement tie	Required operator or gate
T	Primary variable and current OGS VTU output.	Constrained field for the present isothermal workflow.	Thermal residuals are not active; $T = T_0$ fixes thermal gradients and temperature-change terms.
S_l	Secondary variable and current OGS VTU output.	Water-content streams after transformation: NMR, Taupe/TDR, and ERT-derived moisture.	Must be converted to $\theta = nS_l$ or another accepted observable; bound-water and calibration policies remain gates.
n	Material field and current OGS VTU output.	Support for $\theta = nS_l$, not an observed active unknown yet.	Current $n = 0.105$ support must be retained unless NMR/ERT/Taupe policies justify a porosity release.
$\mathbf{K}$ or $e^\top \mathbf{K} e$	Input field, not a state output unless sampled from mesh data.	Pulse-test permeability/transmissibility interpretations.	Interval orientation, support averaging, gas/slip correction, and duplicate-support likelihood policy are required.
$\mathbf{v}_l, \mathbf{q}_l$	Secondary velocity available from OGS; flux can be derived.	Hydraulic diagnostics and possible flow-rate checks if such data are supplied.	Use Darcy velocity, not pore velocity, unless an observation operator explicitly requires division by nS_l .
$\boldsymbol{\sigma}, \boldsymbol{\varepsilon}$, swelling stress	Secondary variables available from the process.	Mechanical plausibility, crack/deformation interpretation, and the prestress gate.	Initial stress and excavation unloading are not active; mechanical conclusions need separate staged-model or post-excavation assumptions.
$\theta = nS_l$	Post-processing product.	NMR and Taupe/TDR water-content comparison; ERT if transformed through a water-content/resistivity relation.	Decide absolute versus bias-corrected versus trend/anomaly comparison and handle bound/interlayer water.
Resistivity or log-resistivity proxy	Observation-model product, not an OGS state variable.	ERT inversion fields.	Requires Archie/clay-specific transform, pore-water conductivity/salinity treatment, surface conduction, anisotropy, support aggregation, and covariance policy.

1.5 Model-facing Question Inventory and Proposed Answers

This section keeps only the questions that still define an unresolved model choice or require external confirmation. Resolved facts are no longer repeated as questions here; they are stated in the preceding prose and in Tables 1.9, 1.8, and 1.12.

Table 1.13: Live model-facing questions after resolved setup facts have been moved into the model text and mapping tables. Questions are shaded light grey and answers are unshaded.

Hydraulic Boundary-Data Provenance

Question. [Q006] Which timestamp and time-origin convention produced the active niche curve $p_E(t)$?

Answer. The active OGS input is clear: the open niche uses `open_niche_seasonal`, while the square boundary is treated as fixed-pressure support. The open query is the source time axis. The provider should confirm the source timestamps, time zone or date convention, interpolation/smoothing, mapping to OGS $t = 0$, and whether the ERT-suggested time shift seen in initial tests also applies to the boundary curve.

Initial State, Excavation, and Prestress

Question. [Q012] Does the initial pressure encode post-excavation drainage near the niche?

Answer. No. The current input has no heterogeneous initial drainage field from RH/suction, direct pressure, nearby boreholes, or excavation history. A measurement-based initialization needs an accepted source field and date before it can replace the uniform value.

Question. [Q013] Does the model simulate staged excavation or gradual unloading?

Answer. No. The computation starts from the already-open 2D geometry, with $\mathbf{u}(\mathbf{x}, 0) = \mathbf{0}$, no active initial stress tensor, and no active top load. The elapsed time from excavation to the model start date is not closed by the checked OGS files.

Question. [Q014] Which prestress magnitude, angle, and excavation reference should be used if prestress is activated?

Answer. Published CD-A modelling sources give stress-magnitude evidence, but no accepted active prestress tensor for this run. The water-content modelling paper reports $\sigma_h = 3$ MPa and $\sigma_v = 7$ MPa [30, Sec. 3]; the GETE modelling paper lists $\sigma_0 = -(5, 5, 7) \times 10^6$ Pa [28, modelling Table 2]. The current XML does not apply either value, and a usable 2D prestress still needs principal directions or rotation angle, component order, sign convention, value choice, and an excavation-reference state.

OGS Inputs and Material-Field Questions

Question. [Q025] Are mechanical parameters fitted now?

Answer. No. Orthotropic elasticity, $\alpha_B = 1$, Bishop exponent 1, and saturation-dependent swelling are active but fixed. Elastic-parameter fitting requires a later internal modelling decision, hydro-mechanical residuals, and a settled prestress/excavation reference state.

Question. [Q027] Is there expert consensus that material fields should evolve in time during the multi-year experiment?

Answer. The current OGS baseline uses time-independent permeability, porosity, retention, and mechanical material fields. Expert input is needed on whether clay sealing, fracture closure/opening, EDZ healing, or other time-dependent material evolution is physically expected strongly enough to justify adding fitted time-dependent fields rather than explaining the trends by boundary forcing, saturation state, or spatial heterogeneity.

Model Time and Output Operators

Question. [Q032] How are output times tied to observations or boundary curves?

Answer. The run starts at $t = 0$, identified by comments as 18 September 2019, and ends at 1.4×10^8 s. Measurements require an explicit date-to-model-time conversion, tolerance, and interpolation or matching rule. This gate is now especially important because initial ERT tests suggest a possible time-axis shift.

Material and Structural Questions

Question. [Q042] Does the same bedding angle control mechanics, permeability, and electrical anisotropy?

Answer. The cited structural evidence supports bedding orientation as important context for permeability candidates, but it does not prove that Darcy, mechanical, electrical, and prestress orientations must share one hard angle. Tensor rotations should state their bedding-angle, prestress-angle, and coordinate conventions explicitly rather than hiding the choices inside component values.

Question. [Q044] How should faults, cracks, and EDZ features be classified here?

Answer. They are not active discrete OGS geometry or separate material classes in this chapter. They can become structural priors, masks, equivalent fields, or scenarios only after their source geometry, scale, aperture meaning, and 3D-to-2D projection are defined.

Chapter 2

Measurements and OGS Observation Links

2.1 Purpose and Evidence Boundary

This chapter records the measurement side of the CD-A Niche 4 modelling problem. The purpose is not to turn every available dataset into a likelihood term. The purpose is to state, stream by stream, what is measured, where and when it is measured, which unit and reference convention are used, and how the value can enter the OGS workflow as an input, output residual, material-field prior, diagnostic, or caveat. This is why each measurement stream first states the measured quantity and support before assigning any model link, residual role, diagnostic role, or gate.

The primary evidence for the plotted data and row summaries is the set of email- and TeamBeam-provided source files named in the bibliography. We use the processed observation tables and generated measurement figures copied into this paper as traceable reductions of those files. The processed-observation layer is the authority for row counts, time ranges, coordinate status, and current activation gates [5, source filenames and transfer context]. Public internet sources were checked only where the provided source catalogue lacks residual-ready numeric exports. They confirm the broader CD/CD-A monitoring context, including pressure, extensometer, crackmeter, Taupe/TDR, RH, water-content, and resistivity streams, but they do not replace the missing project-specific Geoscope, pressure, or crack-opening time series [26, abstract and Sec. 2] [10, Secs. 2.4 and 3.2].

The geometry rule is the same for all streams. A measurement is not an OGS datum until its support is mapped into the current two-dimensional model frame. Point measurements need coordinates and a sampling rule. Borehole interval measurements need segment geometry and interval weights. ERT needs a mesh-to-mesh observation operator. Taupe/TDR needs line-band support. HM monitoring needs a reference frame, instrument sign convention, and a 3D-to-2D reduction policy before it can be weighted.

The model-facing scope in this report is the open-niche side. In the provided source-layout geometry the open twin is the negative- x niche `Open_twin`; after the processed OGS/ERT projection this side is the near-origin cluster and the ERT loop. The measurement-coordinate rows around $x \approx 12\text{--}19\text{ m}$ are the opposite/closed-side context and are not part of the current open-niche comparison unless a section explicitly marks them as qualitative source context [5, source filenames and support reductions].

Table 2.1: Measurement stream roles in the current OGS workflow.

Stream	Current status	Model link	Main gate or use
Permeability pulse tests	Active material likelihood	Direct interval-scale observation of intrinsic permeability $\mathbf{K}(\mathbf{x})$.	Usable for 75 mapped open-twin BCD-A32/A33 rows; support conflicts and older endpoint geometry remain gates.

Table 2.1: Measurement stream roles in the current OGS workflow (continued).

Stream	Current status	Model link	Main gate or use
NMR water content	Active but provisional state likelihood	Observation of water-content state, compared to $\theta = nS_l$ after an observation policy.	Niche 4/open-side rows are relevant here; raw absolute water content is caveated by bound/interlayer water; trend/anomaly is the preferred next residual.
ERT	Diagnostic state field	Forward operator from S_l to water content to bulk resistivity to projected ERT resistivity.	The source archive and electrode loop used here are the open-niche ERT files; the transform, support mask, and covariance remain gates before likelihood use.
Taupe/TDR	Diagnostic trend field	EDZ-band trend operator tied to $\theta = nS_l$, or to dielectric/water-content calibration if confirmed.	Absolute unit and calibration are not confirmed; A3/A4 are the open-side supports, while A7/A8 remain closed-side source context.
RH/suction	Boundary-condition check	Kelvin conversion to pressure or suction candidate for the open-niche boundary $p_E(t)$.	Active curve provenance, filtering, time origin, and uncertainty remain open.
Direct pressure sensors	Blocked future pressure residual	Candidate samples of $p_l(\mathbf{x}, t)$ or hydraulic head.	Mini-piezometer numeric time series, units, reference convention, and quality flags are missing.
Fault/fracture geometry	Structural prior or scenario field	Candidate prior for permeability, porosity, EDZ masks, or 2D projection scenarios.	3D geometry, measured aperture/effect, timing, and projection policy are incomplete for hard use.
Other HM monitoring	Blocked or qualitative HM validation	Candidate displacement, strain, crack opening, levelling, scan-difference, or pressure residuals.	Layout and levelling summaries exist; hard residuals need numeric exports, timestamps, units, reference zeros, and uncertainties.

Uncertainty is stream-specific in the current evidence set. Direct permeability uses a conservative 0.5 log10-unit policy and duplicate-aware weighting. NMR has reported confidence intervals and a floor, but absolute values remain biased by bound/interlayer water. ERT, Taupe/TDR, RH/suction, direct pressure, fracture geometry, and HM streams retain gates until their covariance, calibration, support, and reference conventions are supplied. For practical weighting, only direct permeability and the ERT inversion reports currently carry explicit numeric likelihood scales in this paper: direct permeability uses the 0.5 log10-unit policy, and the ERT inversion reports use log10-resistivity residual scales such as $\sigma_\rho = 0.18$. The other streams remain diagnostic or gated until their own noise model is supplied.

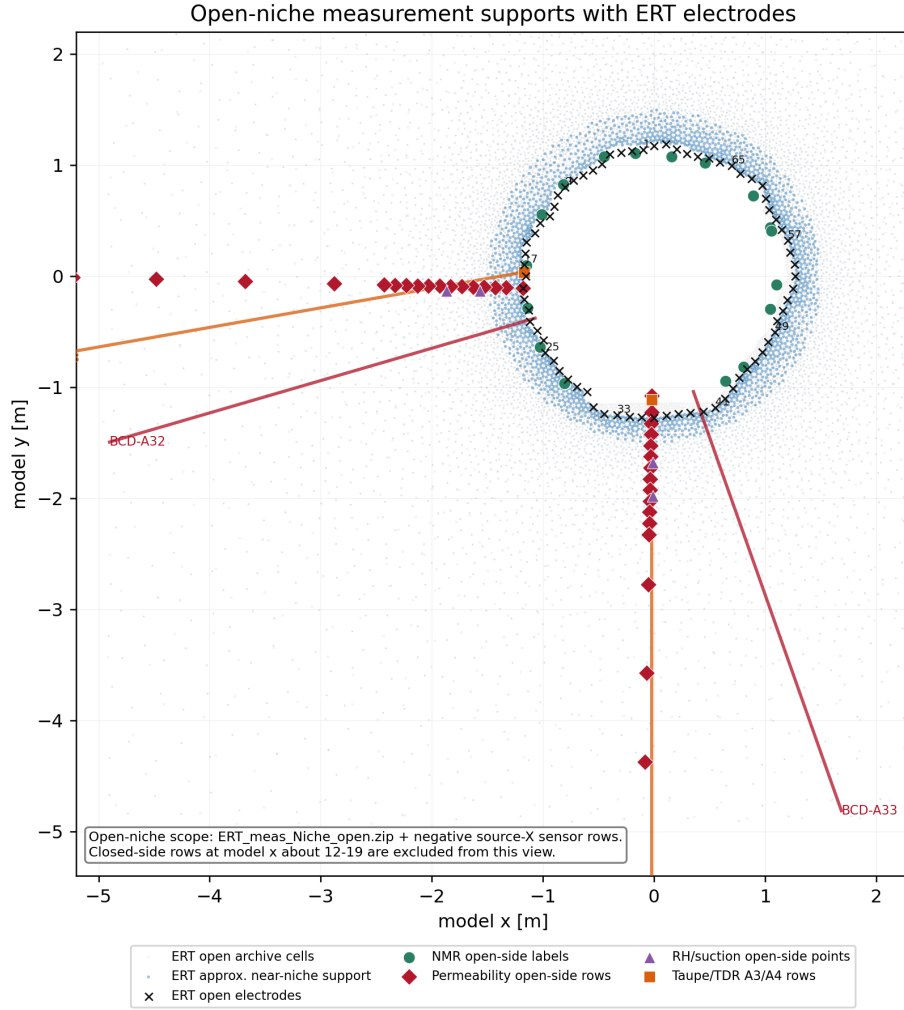


Figure 2.1: Open-niche measurement supports in the current OGS domain. The plot shows the open ERT electrode loop, projected ERT support cells, NMR labels, Taupe/TDR A3/A4 supports, RH/suction points, and open-side permeability pulse-test intervals. Closed-side rows at $x \approx 12-19$ m are excluded from the current model-facing scope.

2.2 Direct Permeability Pulse Tests

Meaning. The permeability stream records interpreted pulse-test permeability values, transmissibility values, and raw pressure-decay traces. The source method is a modified double-piston packer test with a nominal 10 cm interval and nitrogen injection. Therefore the model-facing quantity is a gas-test interpretation of intrinsic permeability, not hydraulic conductivity, not liquid relative permeability, and not a direct tensor component. Gas apparent permeability has its own pressure dependence, so the liquid-equivalent intrinsic value must remain separated from the OGS relative-permeability factor [12, abstract] [15, abstract and Secs. 2.2, 3.2].

Position in the domain. The current active supports are the mapped BCD-A32 and BCD-A33 open-twin intervals inside the Niche 4 OGS mesh. Figure 2.2 shows these supports as borehole-line and cell-sampling locations. Older rows from BCD-A24, BCD-A25, BCD-A26, BCD-A27, and BFM-D19 remain visible in the catalogue, but they do not have accepted endpoint geometry or current-domain support for the first direct fit.

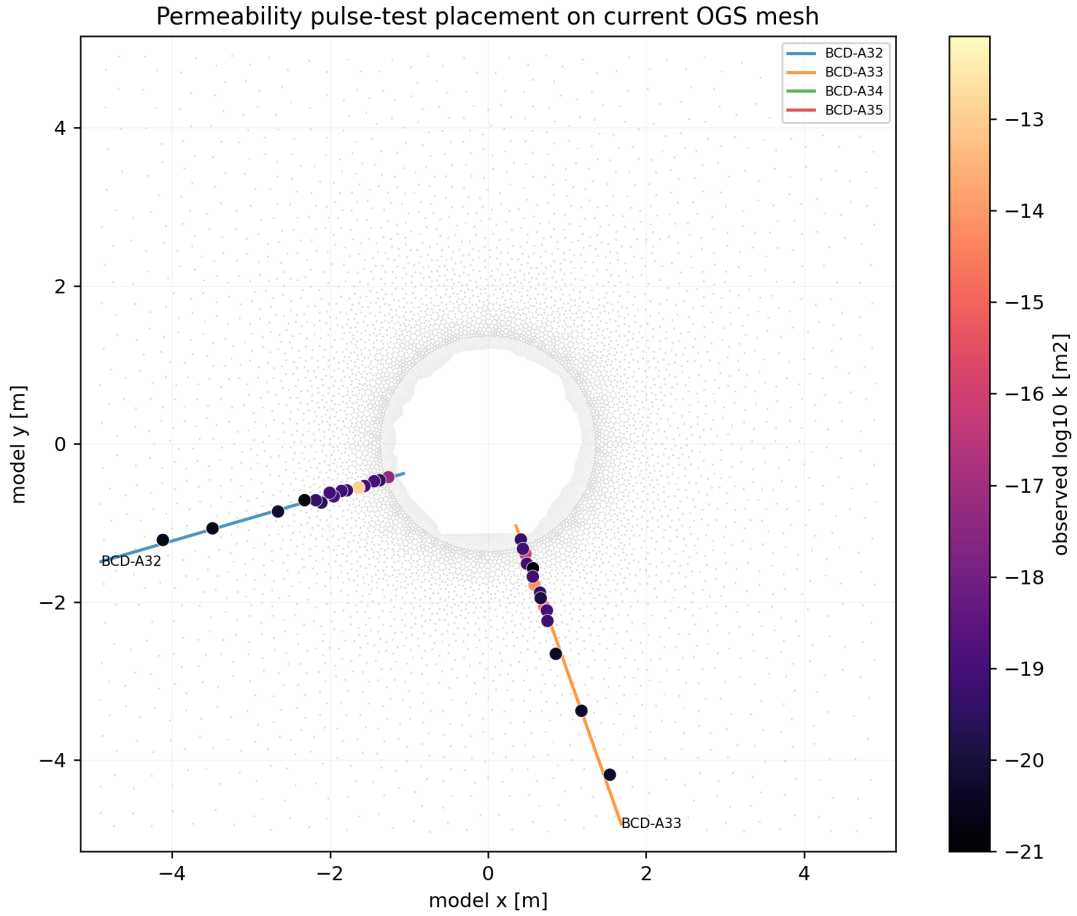


Figure 2.2: Location of active direct permeability supports on the current OGS mesh. The plotted supports are the BCD-A32 and BCD-A33 pulse-test intervals currently usable as direct constraints on $\mathbf{K}(\mathbf{x})$.

Time of measurement. The processed table contains 204 interpreted rows and 6687 pressure-decay rows. The current objective uses the 2024 BCD-A32/A33 rows as quasi-static material evidence. The 2021 rows are retained as open/closed and historical context, but they are not part of the current active objective unless their support policy is reopened.

Measured value, units, and model link. The interpreted value is permeability in m^2 , used in $\log_{10}$ space. The active usable rows span roughly $-21.0 \leq \log_{10}(k/\text{m}^2) \leq -12.10$. The model link is direct: each

row constrains an interval-weighted directional response of the intrinsic permeability tensor,

$$r_i^k := \log_{10} k_{i,\text{pred}} - \log_{10} k_{i,\text{obs}}, \quad k_{i,\text{pred}} = \sum_{c \in \mathcal{C}_i} w_{ic} \mathbf{e}_i^\top \mathbf{K}_c \mathbf{e}_i.$$

Here $\mathcal{C}_i$ is the set of OGS cells sampled by row i , w_{ic} are support weights, $\mathbf{e}_i$ is the interval or borehole-axis unit direction, and $\mathbf{K}_c$ is the cell-wise intrinsic-permeability tensor. If this comparison is used in a posterior, the likelihood term is built from the uncertainty-normalized mismatch $(\log_{10} k_{i,\text{pred}} - \log_{10} k_{i,\text{obs}})/\sigma_i^k$, not from the plotted permeability value alone. This residual acts on the material field used by Darcy’s law, so permeability changes also affect pressure diffusion, saturation evolution, NMR water-content predictions, ERT resistivity predictions, and Taupe/TDR trends.

Uncertainty and reliability. The first-pass direct residual uses a 0.5 log10-unit standard deviation and duplicate-aware weights. This is a policy choice consistent with the order of the experimental/evaluation uncertainty, not a proof that every support conflict is resolved. Same-support rows can differ by several log10 units, and some tests appear to jump rapidly between the lowest and highest reported values. Those oscillations should be treated first as a quality-control and interpretation gate, not as independent high-frequency material evolution. Outlier policy, support aggregation, endpoint recovery, raw pressure-decay interpretation, and repeat-test grouping remain active gates.

Current computational use.

With the available source data now, the defensible use is a direct material-field likelihood on the mapped positive BCD-A32/A33 rows. The nitrogen pulse-test value is treated as a noisy scalar observation of the liquid-equivalent intrinsic permeability used by Darcy flow. The observation operator is the interval average of $\mathbf{e}^\top \mathbf{K} \mathbf{e}$ over the mapped support, evaluated in log10 space. This is the best current transfer from a gas pulse test to OGS $\mathbf{K}$: keep the reported value as intrinsic permeability in m^2 , keep the gas/slip interpretation caveat visible, and absorb unresolved gas-to-liquid and support interpretation errors through a conservative log-space uncertainty and duplicate-aware weighting [12, abstract] [15, abstract and Secs. 2.2, 3.2] [5, source files used for the generated support and row summaries].

This stream directly constrains only $\mathbf{K}(\mathbf{x})$ or its varied parameterization. It does not independently constrain pressure, saturation, porosity, retention, or relative permeability. If the unresolved gas correction, interval support, tensor direction, or duplicate policy is wrong, an inverse run will express that error as false spatial heterogeneity in $\mathbf{K}$.

Grounding questions.

Table 2.2: Grounding questions for direct permeability pulse tests.

Question. [Q045]	How should tests that rapidly oscillate between the lowest and highest interpreted permeability be understood?
Answer.	Treat them as an interpretation and reliability gate before using them as hard likelihood terms. The provider should clarify whether the oscillations are raw pressure-decay fitting instability, leakage or packer effects, gas-slip correction, support duplication, reprocessing duplicates, or real time evolution. Until then, use robust or duplicate-aware weighting, group repeated supports, and avoid reading the extremes as independent fine-scale $\mathbf{K}$ structure.

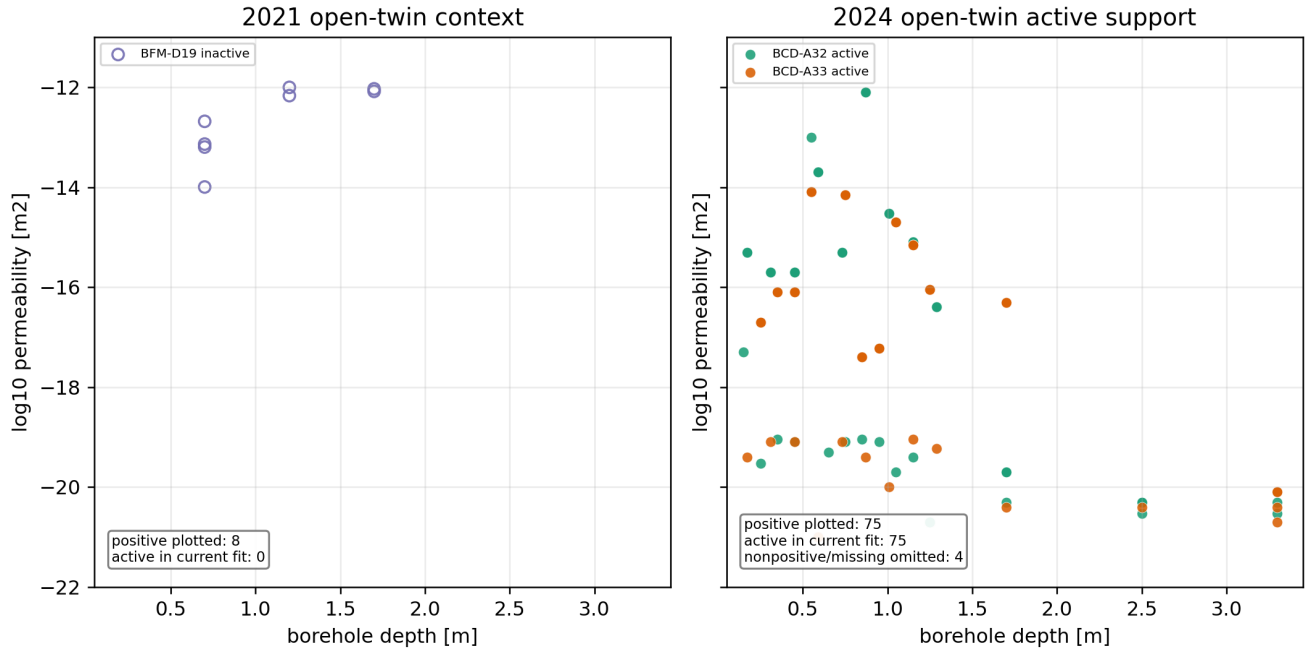


Figure 2.3: Open-twin interpreted permeability rows by campaign year. The 2024 BCD-A32/A33 values form the active support; earlier rows remain context until geometry and support policy are accepted.

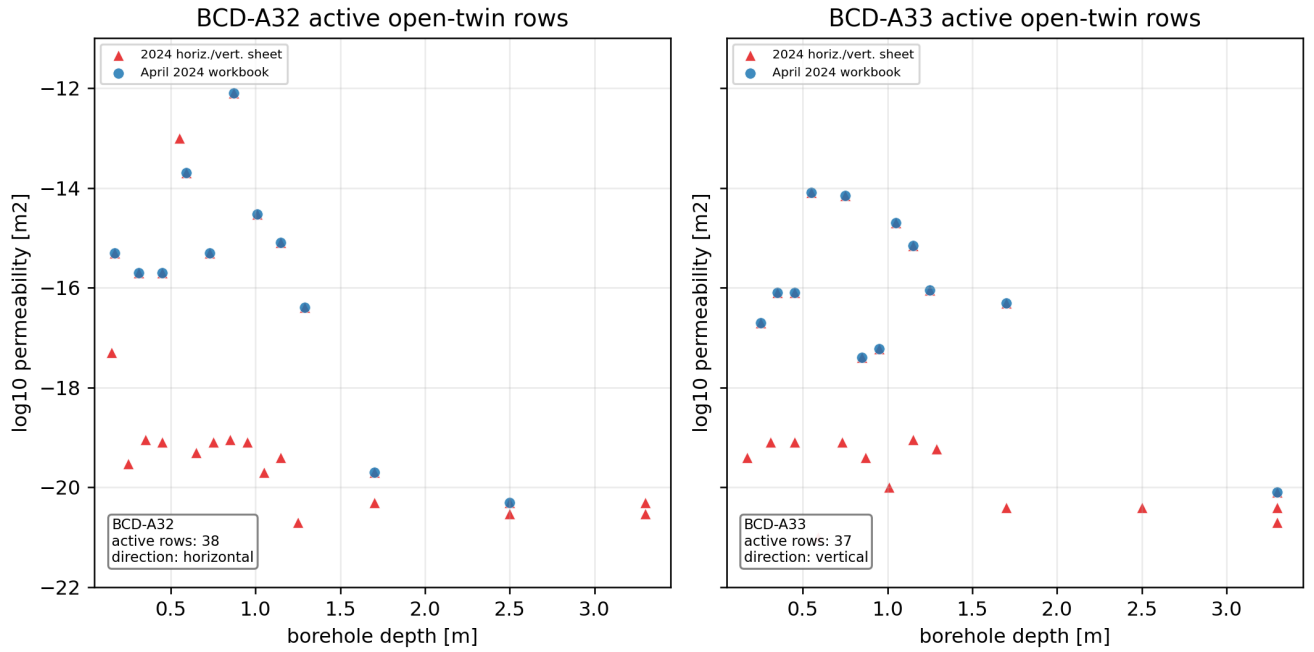


Figure 2.4: Active open-twin permeability rows by borehole. BCD-A32 and BCD-A33 provide the current horizontal and vertical support families for the direct material-field likelihood.

2.3 NMR Water-Content Measurements

Meaning. NMR is the strongest provided water-content evidence, but it is not identical to OGS liquid saturation. The weekly and seasonal source tables report volumetric water content in vol.% and relaxation-time information. In claystone the NMR signal can include mobile pore water and bound or interlayer water. The model-side quantity $\theta = nS_l$ is therefore only the mobile-water proxy unless a bound-water observation model is added [30, Sec. 4.2] [9, Secs. 1 and 2.5] [8, single-sided NMR technical slides].

Position in the domain. The processed NMR labels are mapped into the two-dimensional OGS frame when coordinates exist. Figure 2.5 shows the active and contextual support positions. Niche 4 support is relevant for the current model. Niche 3 campaign rows are retained as experimental context, not as direct observations of the present slice.

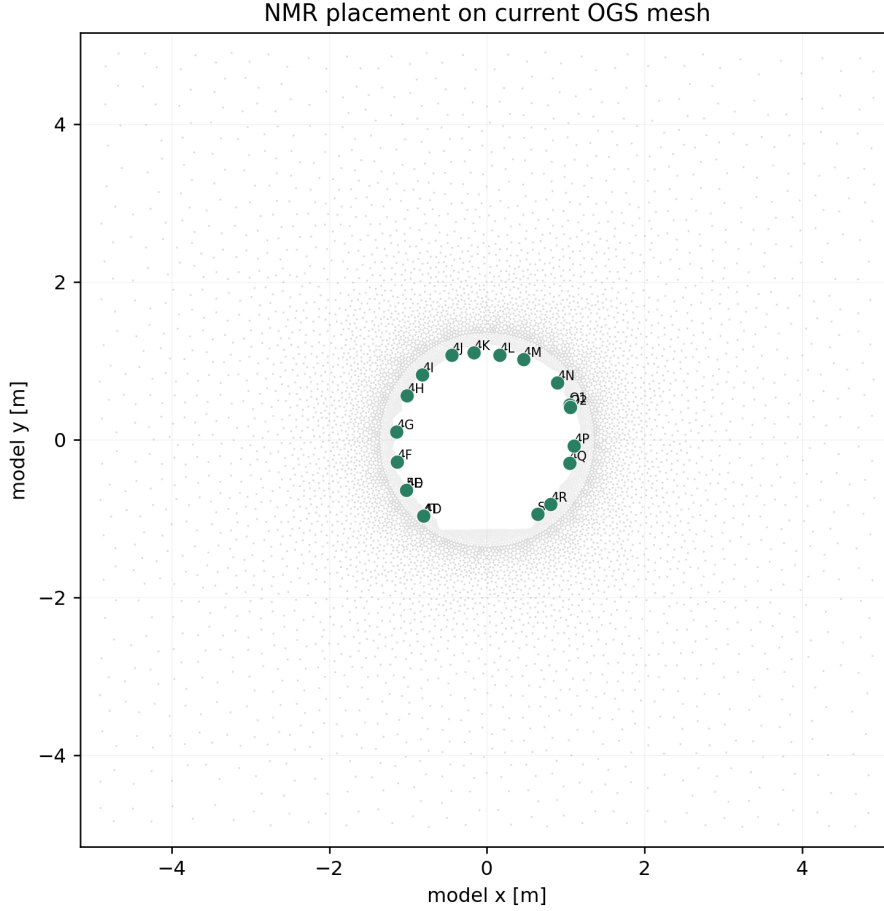


Figure 2.5: NMR support positions in the OGS model frame. These labels define where $\theta = nS_l$ is sampled from OGS outputs for raw and trend/anomaly comparisons.

Time of measurement. The processed NMR layer contains 170 weekly rows and 294 seasonal profile rows. The weekly 4E/4S monitoring covers 2021-10-06 to 2025-09-09, while the seasonal archive contains campaign profiles for repeated Niche 3 and Niche 4 measurement dates. The row-level table keeps caveats such as the 2025 4E detuning period.

Measured value, units, and model link. The measured water content is reported in vol.%; in the model we use the fraction $\theta_{\text{obs}} = \text{vol.\%/100}$. The natural OGS comparison is

$$\theta_{\text{NMR}}^{\text{obs}} = \theta_{\text{mobile}} + \theta_{\text{bound}} + \epsilon_{\text{NMR}}, \quad \theta_{\text{mobile}}^{\text{model}} \approx nS_l.$$

Here θ_{bound} represents NMR-visible bound or interlayer water not carried as an OGS mobile-liquid state, and ϵ_{NMR} represents measurement and observation-model error. The active objective uses raw absolute θ

on the accepted mapped subset. The preferred next policy is a within-label anomaly residual, because it removes constant station or bound-water offsets before comparing model and measurement changes.

Uncertainty and reliability. The bound-water check shows why absolute NMR must remain provisional: with fixed $n = 0.105$, 315 of 464 NMR target rows exceed the maximum possible value of nS_l if interpreted as mobile water. In the usable current-mesh subset, 162 of 287 rows exceed the same bound. Reported confidence intervals and an uncertainty floor are useful, but they do not solve the semantic difference between total NMR-visible water and mobile OGS liquid.

Current computational use.

With the available source data now, NMR can be used as a provisional water-content-state comparison, preferably as a within-label trend/anomaly likelihood rather than a raw absolute likelihood. The current OGS state supplies $\theta_{\text{model}} = nS_l$. The NMR value should be treated as total NMR-visible water unless the source processing proves a mobile/free-water cut. The safest current observation model is therefore

$$\theta_{\text{NMR}}^{\text{obs}} = nS_l + b_{\text{label}} + b_{\text{campaign}} + \epsilon,$$

or the equivalent anomaly model after subtracting stable label/campaign offsets. This matches the provided NMR evidence that many rows exceed the fixed OGS porosity and the cited clay/NMR caveat that NMR-visible water can include bound or interlayer water [30, Sec. 4.2] [9, Secs. 1 and 2.5] [5, NMR source files used for the generated row summaries].

The inverse role is to constrain pressure/saturation evolution after the boundary and permeability field drive S_l . NMR alone cannot separate porosity, retention, mobile fraction, and bound-water offset. Releasing those at the same time without priors would let the inverse problem fit the same water-content residual in several physically different ways.

Grounding questions.

Table 2.3: Grounding questions for NMR water-content measurements.

Question. [Q050]	What exactly does the supplied NMR value measure?
Answer.	Treat it as total NMR-visible volumetric water content for now; provider confirmation is needed for any mobile/free-water interpretation.
Question. [Q051]	How should NMR be linked to OGS output now?
Answer.	Compare to $\theta = nS_l$ only through a bias or anomaly observation model, not as raw mobile water everywhere.
Question. [Q052]	Can bound/interlayer water be anchored simply?
Answer.	Yes, if collaborators confirm that the bound-water offset is approximately constant by label, material class, depth, or campaign; then it can be a nuisance offset rather than an OGS state.
Question. [Q053]	Could bound water be post-processed from OGS saturation?
Answer.	Only if a supported relation is supplied; otherwise it should remain a constant or hierarchical observation offset, not a fitted material field.
Question. [Q054]	What would fully ground NMR residuals?
Answer.	Source units, T2/mobile-water cut policy, calibration and detuning flags, label/campaign offsets, support coordinates, confidence-interval conversion, and a rule for rows where $\theta_{\text{obs}} > n$.

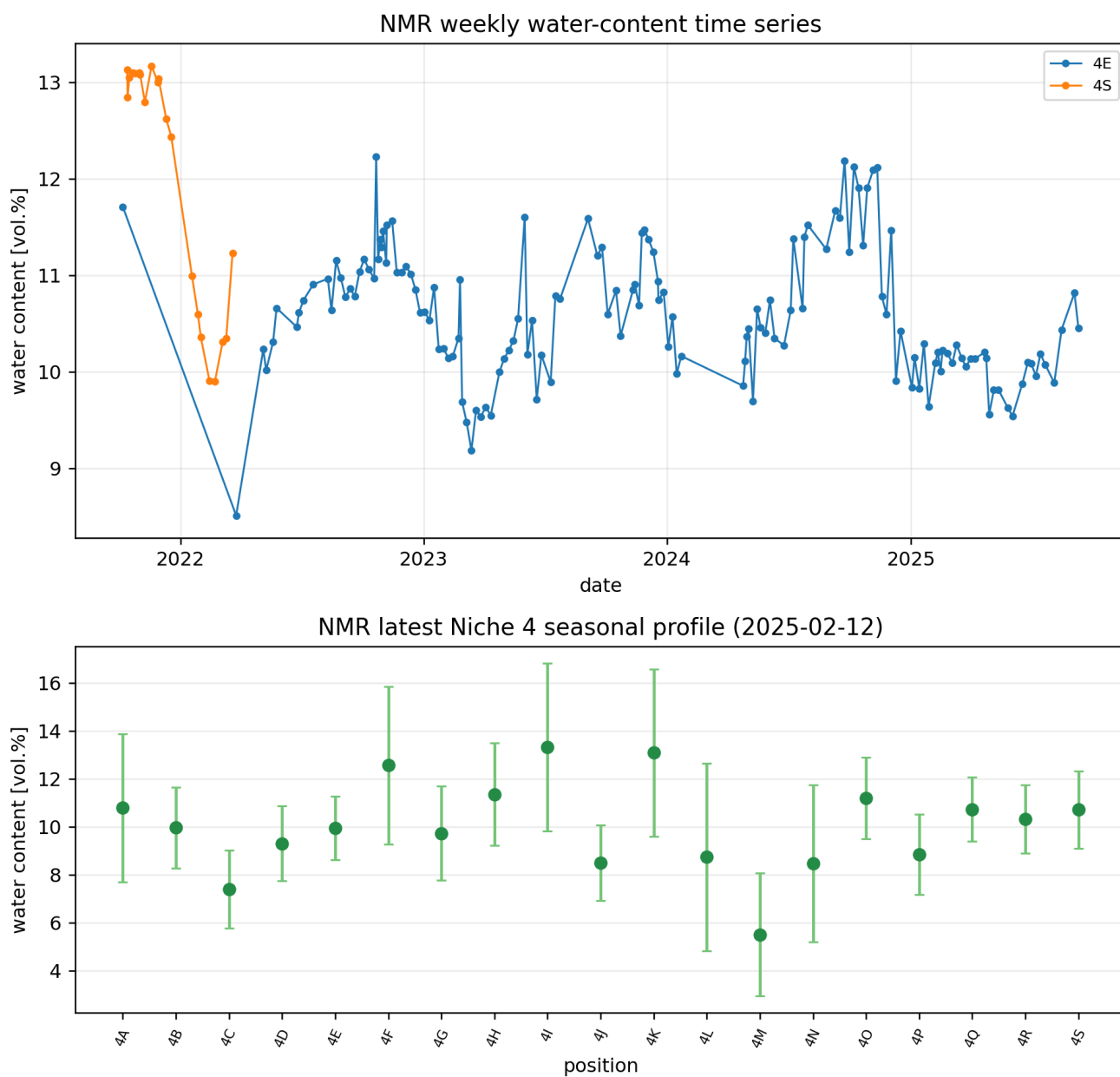


Figure 2.6: NMR measured-data visualisation. The upper panel shows weekly 4E/4S water-content histories. The lower panel shows a seasonal Niche 4 profile with confidence intervals. Values are total NMR-visible water content in vol.%.

2.4 ERT Resistivity Measurements

Meaning. ERT measures an electrical response that is inverted to a resistivity field. It is not a direct measurement of saturation, pressure, or permeability. The physically defensible model link is a forward observation operator: OGS predicts pressure and saturation; saturation is converted to a water-content proxy; water content is converted through the current empirical resistivity relation; the result is projected to the ERT support. Classical Archie-type relations motivate the structure, but claystone requires CD-A-specific calibration because surface conductivity, bound water, fractures, and anisotropy can contribute to electrical conduction [2, pp. 54–56] [20, abstract and Secs. 2–4] [30, Secs. 4.1–4.3].

Position in the domain. The current comparison uses the open-niche ERT archive and the open electrode example file. That archive contains a reference inverted mesh with 5966 triangular cells and 3126 points. The provisional projection transforms raw ERT coordinates by $x_{\text{model}} = x_{\text{ERT}}$ and $y_{\text{model}} = z_{\text{ERT}} + 500 \text{ m}$. It maps 4676 cells inside an OGS cell, with 2035 cells both inside an OGS cell and inside the approximate near-niche support. The closed-electrode example is retained as source context, but it is not used for the current open-niche OGS comparison.

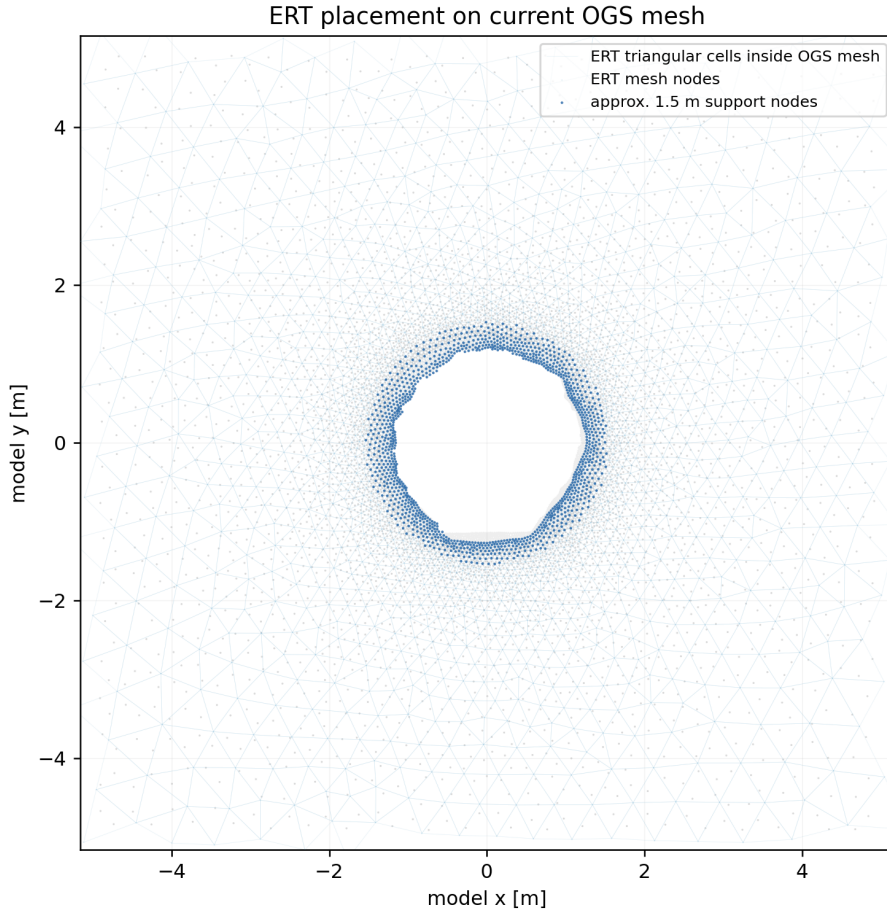


Figure 2.7: Projected ERT mesh support in the OGS frame. The darker subset indicates the current approximate near-niche support used for diagnostics before the exact support mask is accepted.

Time of measurement. The processed ERT inventory contains 1691 timestep entries from 2019-11-01 to 2024-12-31. Of these entries, 1675 have matching VTK fields and 16 are missing a matching VTK member. The archive therefore has dense temporal information, but it must be matched to OGS output times through an explicit tolerance.

Measured value, units, and model link. The comparison should be made in log-resistivity, for example

$$S_l(\mathbf{x}, t) \longrightarrow \theta(\mathbf{x}, t) = n S_l(\mathbf{x}, t) \longrightarrow \rho_{\Omega_m}(\mathbf{x}, t) = 1.108 \theta(\mathbf{x}, t)^{-1.58} \longrightarrow \mathcal{O}_{\text{ERT}}[\rho_{\Omega_m}],$$

followed by

$$r^{\text{ERT}} := \log_{10} \rho_{\text{pred}} - \log_{10} \rho_{\text{ERT}}.$$

Here $\rho_{\Omega\text{m}}$ denotes electrical resistivity in Ωm , and $\mathcal{O}_{\text{ERT}}$ denotes the projection or aggregation from the OGS field to the inverted ERT support. The empirical relation is the current provisional default over the calibrated range $0.01 \leq \theta \leq 0.18$. It should not be interpreted as a universal Opalinus-Clay water-content law.

Uncertainty and reliability. ERT remains diagnostic. The unresolved gates are the exact coordinate transform, the exact tunnel/niche support mask, the 35 cm rock-depth convention, and the uncertainty/correlation model. Without a covariance or aggregation rule, dense neighbouring inversion cells would overweight one physical field image. The processed VTK products are enough for diagnostics, but raw ERT measurements are needed for faster controlled tests, covariance checks, bad-electrode screening, and reprocessing. If the full archive is too large to transfer first, the practical minimum request is the raw electrode-data package for the first two years starting from the 2019-11-01 survey.

Current computational use.

With the available source data now, ERT should remain a diagnostic field and candidate future likelihood. The best current operator is the explicit post-processing chain

$$S_l \rightarrow \theta = nS_l \rightarrow \rho_{\Omega\text{m}} = 1.108 \theta^{-1.58} \rightarrow \mathcal{O}_{\text{ERT}} \rho \rightarrow \log_{10} \rho,$$

using the provisional coordinate transform and near-niche mask only for diagnostics. This keeps the empirical CD-A water-resistivity relation separate from the OGS hydraulic state, and it prevents the inverted ERT mesh from being mistaken for independent point measurements [2, pp. 54–56] [20, abstract and Secs. 2–4] [30, Secs. 4.1–4.3] [5, ERT source files used for the generated field summaries].

For inversion, ERT mainly constrains the saturation/water-content pattern and therefore tests permeability, boundary forcing, and retention indirectly. Its dense cells share electrodes, inversion regularization, and projection errors. Without aggregation or covariance, ERT would dominate the objective numerically while adding much less independent information than the row count suggests.

Grounding questions.

Table 2.4: Grounding questions for ERT resistivity measurements.

Question. [Q056]	Is the Archie-type water-content-log-resistivity fit global, or only a local empirical relation?
Answer.	Treat $\rho_{\Omega\text{m}} = 1.108 \theta^{-1.58}$ as the current local CD-A empirical fit between water content and log-resistivity, not as a global Archie law. Expert confirmation is needed on its calibrated range, whether $\theta = nS_l$ is the right water-content proxy, and whether separate calibrations are needed for bound water, clay surface conduction, pore-fluid chemistry, fractures, or anisotropy.
Question. [Q057]	Does bound water conduct like free water?
Answer.	Not settled; clay surface conduction and bound/interlayer water are explicit gates before absolute water-content inversion.
Question. [Q058]	Does electrical anisotropy matter?
Answer.	Possibly; bedding, fractures, and permeability anisotropy may affect electrical response, but no accepted electrical-anisotropy operator exists for this workflow.
Question. [Q086]	Can we obtain raw ERT data for all campaigns, or at least the first two years?
Answer.	Request raw electrode measurements, timestamps, array configurations, reciprocal or error estimates, current/voltage data, inversion settings, and bad-electrode flags for all ERT campaigns. If that is too large for the first transfer, request the same package for the initial two-year window beginning with the 2019-11-01 survey, so time-shift and likelihood tests can be run quickly.

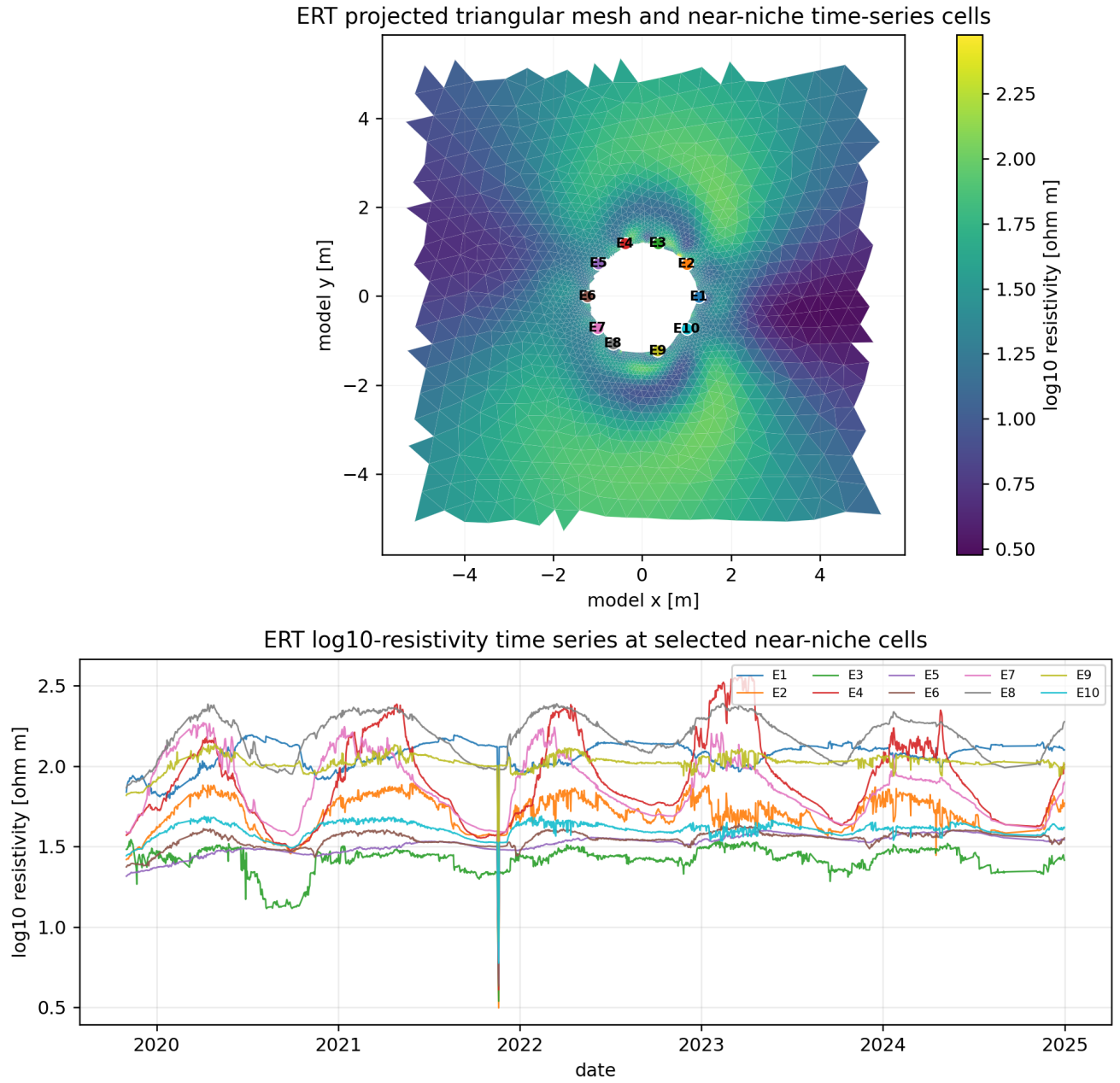


Figure 2.8: ERT measured-data visualisation. The upper panel shows a projected reference log₁₀-resistivity field and selected support cells; the lower panel shows their log₁₀-resistivity histories over matched ERT timesteps.

2.5 Taupe/TDR Measurements

Meaning. The Taupe/TDR workbook provides a dielectric or ARDP-derived water-content proxy. The provided files do not yet prove whether the values are calibrated volumetric water content, apparent relative dielectric permittivity, or another Taupe-specific proxy. For that reason the safe current interpretation is a same-series trend diagnostic, not an absolute saturation residual. General TDR literature supports the link between dielectric response and water content, but the CD-A claystone calibration must be supplied before absolute use [29, Sec. 4.4] [21, review and Secs. 1–2] [13, Taupe/TDR technical slides].

Position in the domain. The source sheets A3, A4, A7, and A8 correspond to Taupe supports and EDZ distance bands. A3 and A4 are mapped into the current OGS frame. A7 and A8 are kept in the workbook and operator tables as closed-side context, but their support maps to the $x \approx 12$ m side and falls outside the present open-niche comparison.

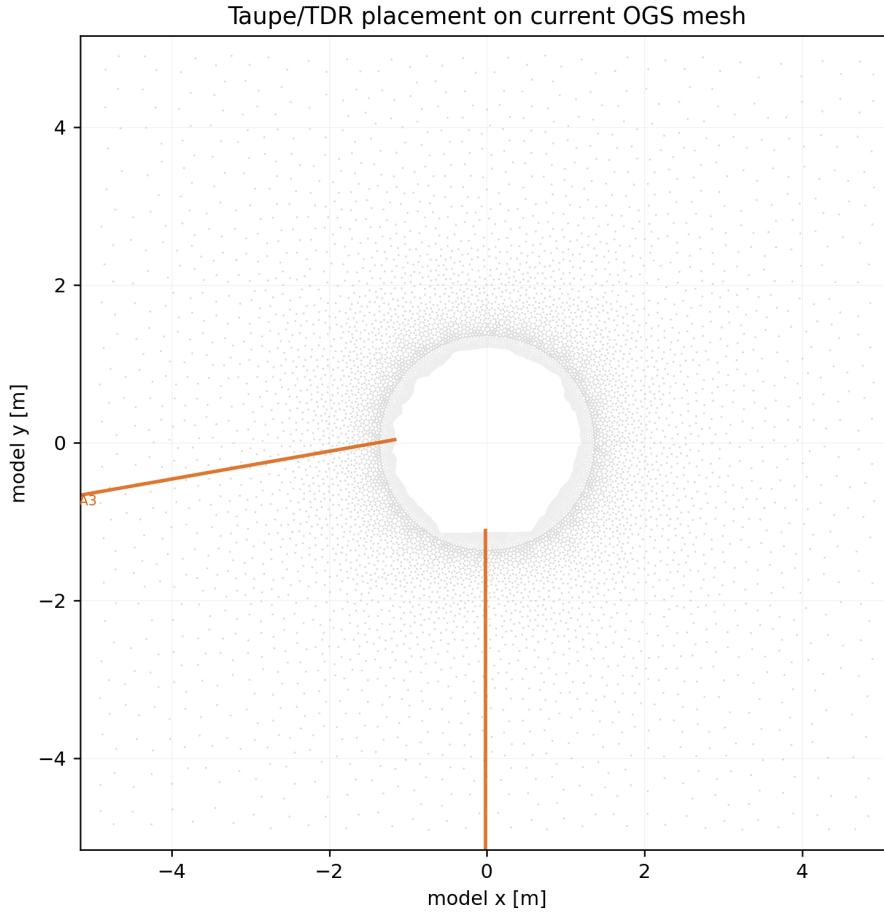


Figure 2.9: Mapped Taupe/TDR line supports. A3 and A4 are the current in-domain supports for trend diagnostics; A7 and A8 remain source-data context.

Time of measurement. The processed Taupe/TDR layer has 5088 sensor-date-band rows from 2019-12-01 to 2025-10-10. The rows are grouped by sensor and EDZ band, including 0–10, 10–20, 20–30, 30–40, 40–50, and aggregate 0–50 cm bands from the niche wall.

Measured value, units, and model link. The workbook values span approximately 9.87 to 11.00 in source units. Until the unit is confirmed, the model comparison is the standardized trend

$$r_i^{\text{Taupe}} := H_i^{\text{Taupe}}(u) - d_i^{\text{Taupe}}, \quad \theta_{\text{model}} = nS_l,$$

where H_i^{Taupe} is the sensor/band observation operator applied to the OGS state and d_i^{Taupe} is the measured Taupe/TDR value after the accepted baseline or calibration. In the current diagnostic mode, H_i^{Taupe} and

d_i^{Taupe} are standardized within sensor and EDZ band; if a calibration is supplied, the same support can become a calibrated direct-value likelihood residual in physical water-content, permittivity, or another stated unit. That residual informs the posterior only after it is paired with an uncertainty, for example through $(H_i^{\text{Taupe}}(u) - d_i^{\text{Taupe}})/\sigma_i^{\text{Taupe}}$.

Uncertainty and reliability. The trend operator is ready as a diagnostic, but calibrated likelihood residual weights are not. The missing items are the workbook unit, calibration equations, baseline convention, sensor and band uncertainties, and whether bound/interlayer water affects the dielectric response in the same way as mobile pore water.

Current computational use.

With the available source data now, Taupe/TDR can be used as a same-sensor, same-band trend diagnostic for mapped A3/A4 supports. The current operator should band-average $\theta_{\text{model}} = nS_l$ over the corresponding EDZ-depth support, subtract a series baseline, standardize within sensor and band, and compare only the timing, sign, and relative amplitude of changes. This avoids pretending that the workbook unit is already an absolute volumetric water content or dielectric permittivity. The literature supports the general TDR water-content link, while the Taupe stream still needs CD-A-specific calibration before absolute use [21, review and Secs. 1–2] [13, Taupe/TDR technical slides] [5, Taupe source workbook and support files].

The inverse role is indirect: Taupe/TDR can help reject permeability or boundary candidates that produce the wrong wetting/drying trend near the EDZ. It cannot yet identify absolute saturation, porosity, retention, or bound-water response because the sensor unit and calibration are still open.

Grounding questions.

Table 2.5: Grounding questions for Taupe/TDR measurements.

Question. [Q060]	What does the Taupe workbook value represent?
Answer.	Unknown from the available source files; treat it as a source-unit dielectric/water-content proxy until calibrated units are confirmed.
Question. [Q061]	How should Taupe/TDR be linked to OGS now?
Answer.	Compare standardized EDZ-band trends against band-averaged $\theta = nS_l$ changes for A3/A4.

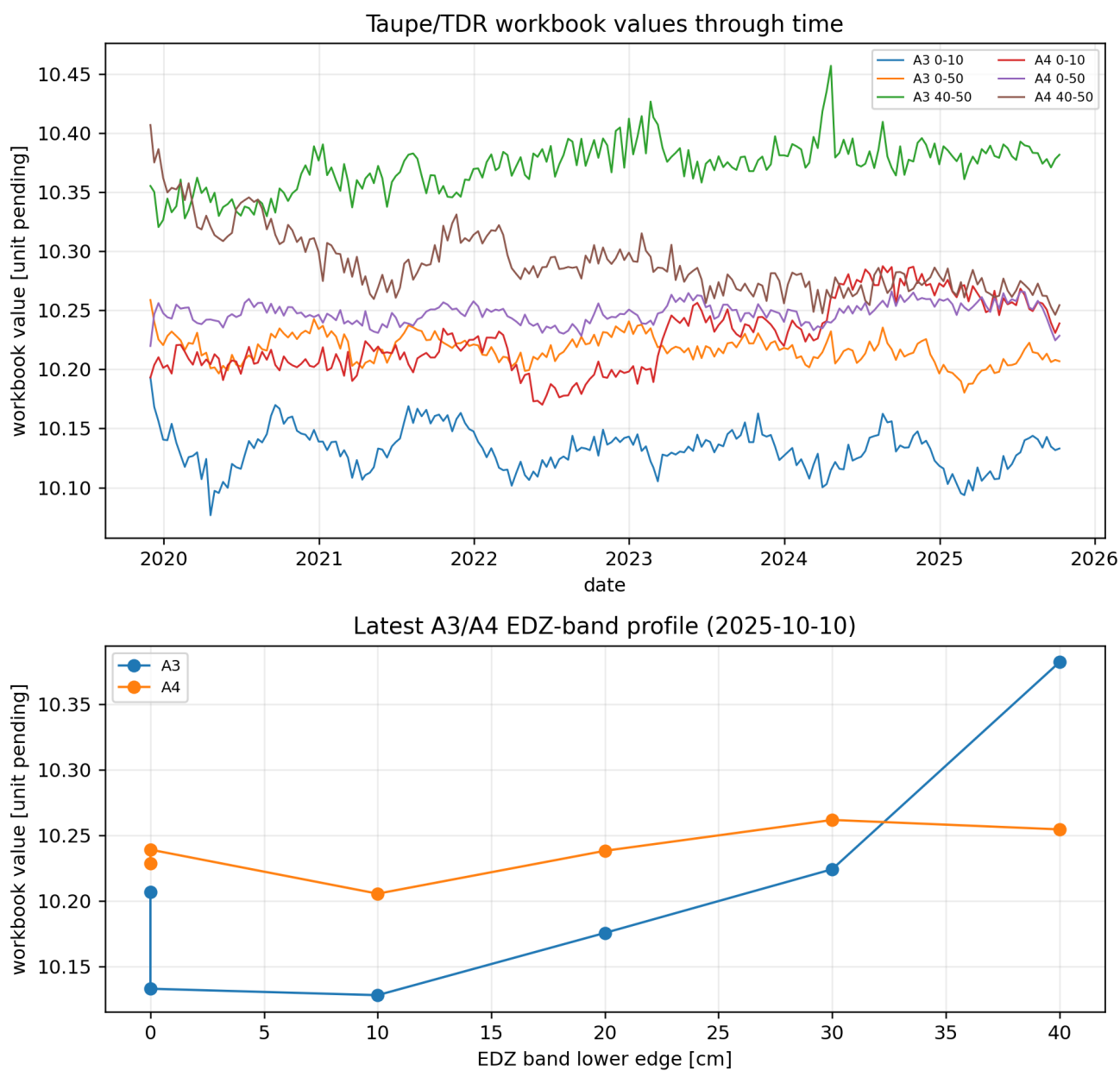


Figure 2.10: Taupe/TDR measured-data visualisation. The upper panel shows representative A3/A4 EDZ-band histories; the lower panel shows a latest EDZ-band profile. Values are retained in source workbook units.

2.6 RH, Suction, and Open-Niche Pressure Forcing

Meaning. The RH workbooks measure relative humidity, not liquid pressure directly. Under a Kelvin-equation convention, RH can be converted to a capillary-pressure or gauge-liquid-pressure candidate. This makes RH primarily a boundary-condition source or check for the open niche. It can later inform retention or boundary uncertainty, but it is not an interior material residual.

Position in the domain. The RH/suction supports are near the open-niche boundary and are mapped as boundary-check points in the OGS frame. Their role is to test whether the time-dependent boundary curve $p_E(t)$ is consistent with humidity-derived suction evidence.

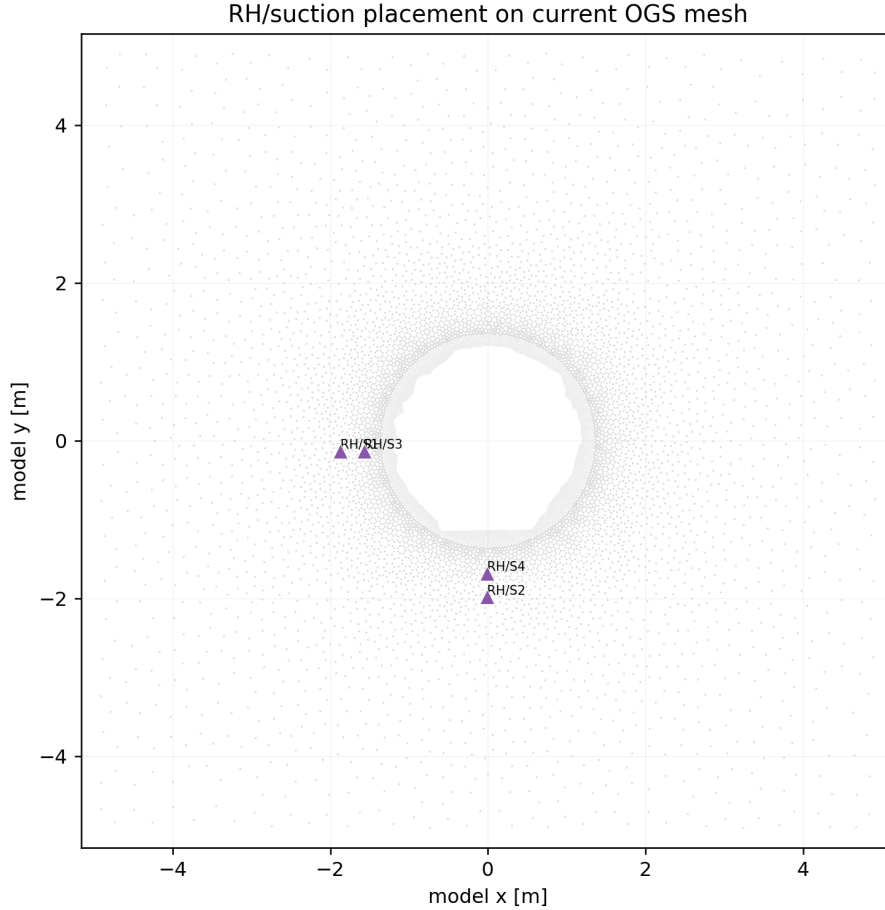


Figure 2.11: Mapped RH/suction support points used to check the open-niche pressure boundary. These are boundary-forcing supports, not cell-wise material observations.

Time of measurement. The processed RH layer contains 4247 daily rows from 2021-12-16 to 2025-09-04 for RH5, RH6, RH7, and RH8. RH5 and RH6 are the cleanest open-twin evidence. RH7 and RH8 contain low outliers near 13% RH and require stronger screening.

Measured value, units, and model link. The source unit is RH in percent. We convert it to a gauge-pressure candidate by

$$p_l^{\text{Kelvin}} = \frac{\rho_l R T}{M_w} \ln(\text{RH}),$$

where RH is a fraction, $T = 298.15$ K, and $\rho_l = 1095$ kg m⁻³. The constants R and M_w are the molar gas constant and molar mass of water used by the Kelvin conversion. Since $\ln(\text{RH}) < 0$, the converted liquid pressure is negative in the same suction convention used by the active open-niche curve. This is consistent with $p_c = -p_l$ in the Richards model [22, pp. 448–452] [30, Sec. 3, Eqs. 1–2].

Uncertainty and reliability. The active OGS curve has 845 rows and spans approximately -53.24 to -5.23 MPa. The RH boundary check compares 2280 rows inside the active curve time range and finds large differences between RH-derived pressure candidates and the active curve. Therefore $p_E(t)$ should remain an existing OGS input, not a verified RH-derived forcing, until the source sensors, filtering, smoothing, time origin, interpolation, and extension policy are reconstructed.

Current computational use.

With the available source data now, RH/suction should be used as boundary-forcing provenance and uncertainty information for the open-niche pressure curve, not as an independent interior residual. The current best transfer is to convert screened RH records to a Kelvin pressure candidate with the stated constants, compare that candidate to the active XML curve $p_E(t)$, and represent the mismatch as a boundary-condition gate or scenario envelope. This is grounded in the active curve values, RH workbook check, and the CD-A water-content/RH formulation where negative liquid pressure represents suction through the chosen pressure convention [22, pp. 448–452] [30, Sec. 3, Eqs. 1–2] [5, RH source workbooks used for the generated boundary checks].

In the inverse problem, uncertain $p_E(t)$ is dangerous because permeability, retention, and boundary suction all control similar drying/wetting transients. If the boundary curve is treated as exact while its provenance is unresolved, the fit can falsely compensate by changing $\mathbf{K}(\mathbf{x})$ or retention parameters.

Grounding questions.

Table 2.6: Grounding questions for RH/suction and pressure-boundary forcing.

Question. [Q067]	Which sensors define the open-niche boundary?
Answer.	RH5/RH6 are the cleanest source candidates, but final use needs explicit sensor selection, filtering, high-RH/low-outlier handling, and time alignment.
Question. [Q087]	Can we obtain the full RH/suction source data from the start of the active $p_E(t)$ curve?
Answer.	Request the complete source data behind the open-niche pressure forcing from its 2019 start: raw RH, temperature, suction or processed-pressure rows, sensor ids, timestamps, screening flags, and the script or table that converts those values into $p_E(t)$. The key provenance question is whether the initial p_E segment in the XML was derived from these measurements, manually prescribed, or extended by a separate preprocessing rule.
Question. [Q069]	What would fully ground the pressure boundary?
Answer.	Source table, sensor selection, smoothing/filtering, interpolation and extrapolation rules, calendar time origin, sign/unit convention, and a covariance or scenario envelope for boundary uncertainty.

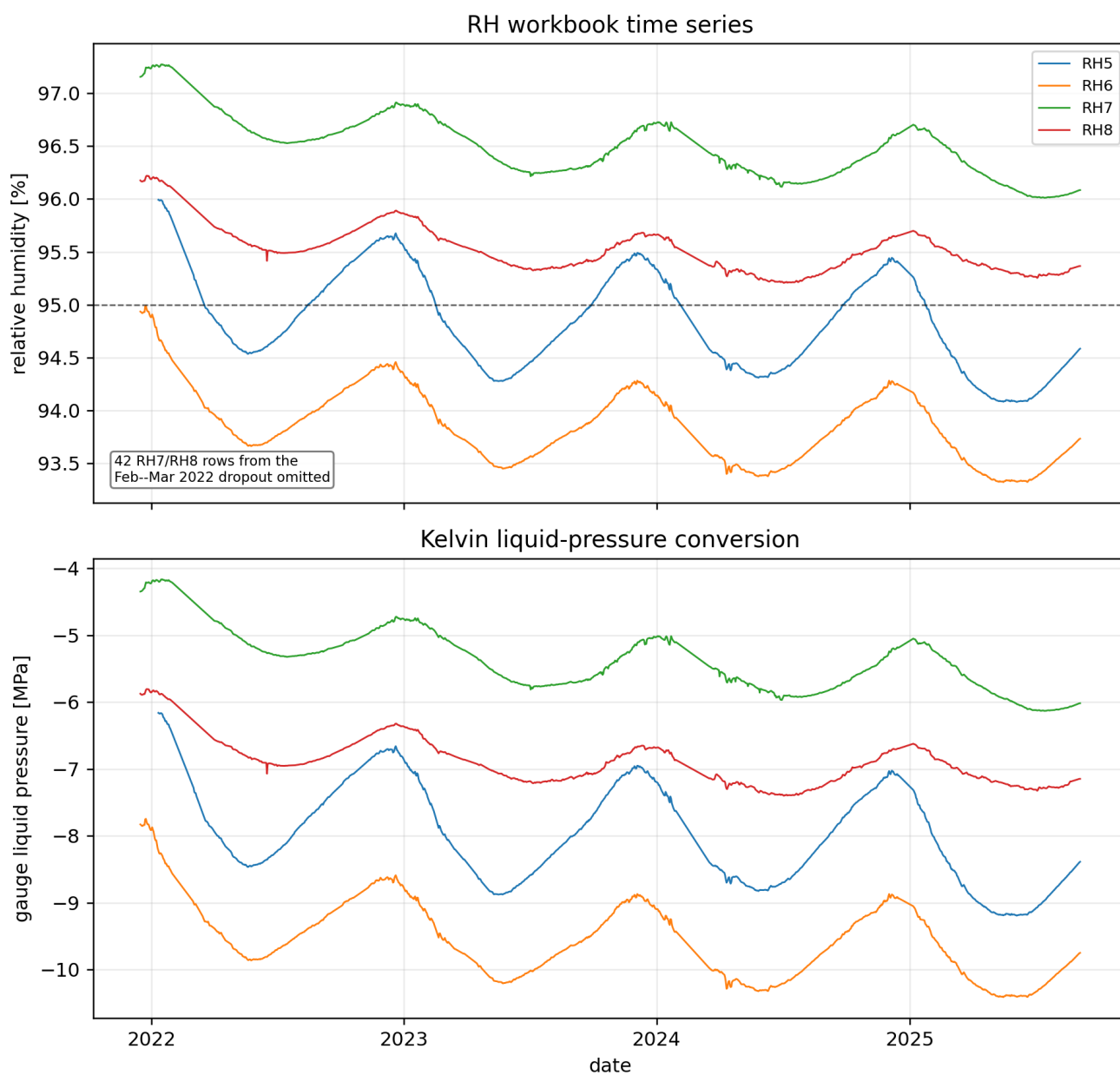


Figure 2.12: RH and Kelvin-converted pressure visualisation. The upper panel shows RH5–RH8 histories after omitting the flagged RH7/RH8 dropout interval; the lower panel shows the corresponding gauge liquid-pressure conversion in MPa.

2.7 Direct Pressure Measurements

Meaning. The direct-pressure stream is the mini-piezometer stream. In the CD-A measurement inventory, mini-piezometers measure pore pressure with unit Pa [29, Table 1]. The VEIS paper confirms that CD-A has borehole sensors for pressure measured by minipiezometers and that sensor databases store sensor names, date/time stamps, and measured values [10, Secs. 2.4 and 3.2]. The 2026 email-provided technical discussion identifies BCD-A28 to BCD-A31 as the current mini-piezometer set and shows their pressure trends in kPa [25, PDF p. 32].

Position in the domain. Figure 2.13 includes the source-slide location inset for A28–A31. That inset is enough to identify the instrument family and open/closed-twin context, but it is not yet an accepted OGS point set. A usable pressure residual still needs individual sensor coordinates or trace supports, sensor elevations/depths, and a rule for projecting each 3D sensor support to the present 2D slice.

Trends in measurements: Mini-Piezometer



Open Twin:

Seasonality and ongoing trend (?) – to be observed further

Closed Twin:

Steady state reached?

(Use of this data for model validation)

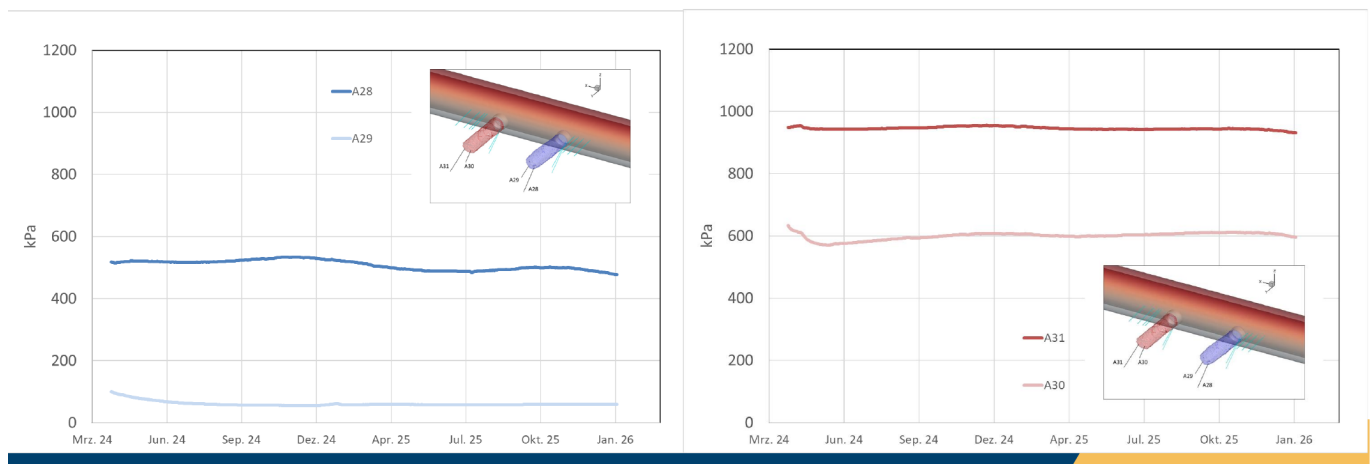


Figure 2.13: Mini-piezometer trend evidence from the 2026 technical discussion slides. The plotted values are in kPa and cover approximately March 2024 to January 2026. The open twin shows seasonality and an ongoing trend; the closed twin is shown only as source context for this open-niche note. This is a source-slide plot, not a machine-readable export or current residual.

Time of measurement. The source plot spans about March 2024 to January 2026. The May 2026 minutes state that BCD-A28 to BCD-A31 were working well, and list pore-pressure measurements as later comparison targets for the modelling workflow [6, PDF p. 4 and p. 8]. The sampled numeric export behind the plot is still absent from the current provided catalogue, so exact timestamps, sampling interval, maintenance gaps, and quality flags remain unavailable.

Measured value, units, and model link. The source plot uses kPa. The public CD-A inventory names the quantity as pore pressure in Pa, but the provided export must still state whether the stored values are absolute pressure, gauge pressure, hydraulic head, or another Geoscope conversion. Once that reference convention is known, the model link is direct: sample $p_l(\mathbf{x}, t)$ at the sensor support and convert pressure unit, elevation head, atmospheric reference, and sensor depth consistently with the model pressure convention.

Uncertainty and reliability. This stream is promising but not residual-ready. The required export must include timestamps, units, sensor ids, coordinates or support ids, zero/reference convention, calibration,

temperature compensation, quality flags, maintenance/failure flags, and raw-file provenance. The 2D model error may be significant if a mini-piezometer samples a local 3D feature or lies outside the effective support of the modeled slice.

Current computational use.

With the available source data now, direct pressure measurements are a blocked but high-value future residual. The source-slide trends prove that the stream exists and has physically meaningful evolution, but the paper does not yet have the machine-readable export needed for fitting. Once that export is available, the operator should be a direct sample or support average of OGS $p_l(\mathbf{x}, t)$, or a hydraulic-head comparison after elevation and reference-pressure conversion. This would directly test whether the far-field pressure, open-niche boundary curve, initial pressure field, and fitted permeability field are mutually consistent [29, Table 1] [10, Secs. 2.4 and 3.2] [25, PDF p. 32] [5, provided pressure/HM source files and generated trend figures].

The most important current modelling use is not to fit the slide image. It is to request the pressure export in a form that closes the pressure-reference ambiguity. Confusing absolute pressure, gauge pressure, hydraulic head, suction, or capillary pressure would shift the residual and can force incorrect changes in boundary pressure, initial pressure, or $\mathbf{K}(\mathbf{x})$.

Grounding questions.

Table 2.7: Grounding questions for direct pressure measurements.

Question. [Q070]	Do direct pore-pressure measurements exist for this model?
Answer.	Yes as a documented mini-piezometer stream and source-slide trend; no residual-ready numeric export is present yet.
Question. [Q071]	How should pressure be linked to OGS?
Answer.	Compare to $p_l(\mathbf{x}, t)$ or hydraulic head only after units, elevation, absolute/gauge/reference convention, and sensor support are known.
Question. [Q073]	What 2D-model caveat is needed?
Answer.	Each sensor may sample a 3D hydraulic path or feature outside the modeled slice; the export must support a defensible 3D-to-2D comparison rule.
Question. [Q088]	Can we obtain the exact data and positions for BCD-A28–A31?
Answer.	Request the machine-readable mini-piezometer time series together with exact sensor positions: coordinates in the source frame, borehole or screened interval, elevation/depth, open- or closed-twin assignment, timestamps, raw pressure units, pressure reference, calibration and maintenance flags, and the accepted 3D-to-2D projection for the current OGS slice.

2.8 Faults, Fractures, and Crack Geometry

Meaning. This section is about large faults, fault zones, and mapped fracture geometry, not about crack-meter time series. The public CD-A characterization paper reports fault planes parallel to bedding as well as non-bedding-parallel fault planes, and summarizes the structural geology near the twin niches in its Fig. 4 [29, Sec. 3 and Fig. 4]. For the OGS workflow, these features are not observations of pressure or displacement by themselves. They are structural information that can affect permeability, porosity, EDZ extent, anisotropy, or prior covariance.

Position in the domain. The provided Tecplot layout contains 22 **Kluft** fracture/crack-geometry zones. Those zones document source-frame structural supports, but they do not yet define which 3D features intersect the current 2D model slice, what effective width should be used, or whether the feature should become a hard material input, a prior, a mask, or a qualitative caveat. The email-provided HERMES input note also states that geological characterization of fractured zones is available for CD-A [3, PDF p. 5].

Time of measurement. The geometry is structural, not a complete time series. A feature can still be used as time-dependent modelling information only if there is measured aperture, displacement, hydraulic effect, or dated structural evolution. Without that, it should be treated as static structural context or as a scenario/prior field.

Measured value, units, and model link. For large faults and fracture zones, the model-facing quantities are 3D coordinates, plane or zone geometry, width/support, aperture or hydraulic effect, and a projection policy. The model link is indirect: a fault may modify $\mathbf{K}(\mathbf{x})$, $n(\mathbf{x})$, EDZ classification, permeability-tensor orientation, or the prior covariance of those fields. It is not a residual against an OGS output unless a measured quantity, unit, time, and observation operator are defined.

Uncertainty and reliability. The main risk is overfitting a two-dimensional permeability field to a three-dimensional unresolved structure. A fault or crack should become a hard input only after the projection policy is explicit: which 3D feature intersects the 2D slice, what width/support is assigned, whether it changes $\mathbf{K}$, n , EDZ classification, or only the prior covariance, and which uncertainty represents projection error.

Current computational use.

With the available source data now, large faults and fracture geometry should enter as structural priors, masks, or scenario definitions rather than residuals. The provided `.dat/Tecplot` geometry gives source-frame structural zones, and the public CD-A characterization supports the relevance of faults and fractured zones, but geometry alone does not define hydraulic aperture, permeability contrast, or a time-dependent material change. The best current use is to construct alternative 2D projection scenarios: no fault effect, a projected EDZ/fault mask, a local permeability-basis support, or a covariance modifier for $\mathbf{K}$ and possibly n or tensor orientation [29, Sec. 3 and Fig. 4] [3, PDF p. 5] [5, geometry source file `VisualisationCDA.dat`].

For inversion, the key discipline is not to turn fault geometry into a fake measurement. A fault can guide where material fields are allowed to vary. It becomes a residual only if a numeric aperture, displacement, pressure response, or other measured value is supplied with units, time, support, and an observation operator.

Grounding questions.

Table 2.8: Grounding questions for large faults and fracture geometry.

Question. [Q075]	What does the large-fault source measure?
Answer.	Currently it is structural geometry; no accepted residual-ready aperture, displacement, or hydraulic-effect table is present.
Question. [Q076]	How should faults link to OGS now?
Answer.	As projected structural priors, EDZ/fault masks, permeability-basis supports, tensor-orientation context, or scenario fields.
Question. [Q077]	Can a fault modify permeability or porosity?
Answer.	Yes as a scenario or prior, but hard modification needs independent evidence from permeability, ERT, pressure, NMR/Taupe trends, or crack/HM data.
Question. [Q078]	How should 3D geometry be reduced to the 2D slice?
Answer.	By an explicit intersection, influence-band, or nearest-distance projection with effective width and uncertainty; otherwise it remains a caveat.
Question. [Q079]	What would fully ground quantitative fault use?
Answer.	Exact source file, coordinates and units, feature identities, 3D-to-2D projection, width/support, target fields ($\mathbf{K}$, n , EDZ), time-dependence assumption, and independent hydraulic or mechanical evidence.

2.9 Geoscope, Crackometer, and Other HM Monitoring

Meaning. The other-HM package covers measurements that can constrain mechanics, pressure, swelling, permeability plausibility, or boundary disturbances: extensometers, crackmeters, laser scans, miniprisma geodesy, convergence points, precision levelling, evapometer or climate context, and Geoscope auxiliary streams. The CD-A inventory identifies extensometers and convergence as displacement measurements, laser scans as niche-profile displacement measurements in micrometres, and mini-piezometers as pore pressure measurements [29, Table 1]. The email-provided HERMES input note frames the same bundle as long-term deformation, humidity, water-content, pore-water-pressure, crack-width, piezometer, extensometer, ERT, TDR, and laser-scan measurements [3, PDF p. 5].

Position in the domain. The source layout contains support geometry for several HM streams, but the generic support plots are not useful paper figures because they do not show measured values, timestamps, units, or OGS-ready support. Table 2.9 therefore records the useful status directly. Before fitting, each stream needs a mapping from source frame to the current 2D model frame and, where relevant, a 3D-to-2D comparison rule.

Table 2.9: Other-HM measurement status in the current source catalogue.

Stream	What it measures	Current source evidence	Model link and status
Extensometer	Displacement or strain along instrument geometry.	Geoscope/minutes evidence; BCD-A9/B10 closed-niche failure since September 2025.	$\mathbf{u}$ or strain diagnostic; hard residual blocked until time series, geometry, sign, and quality flags are supplied.
Crackmeter	Crack aperture or local relative displacement.	Source trend plot and minutes: one closed-twin trend and open-twin seasonal variation.	$\mathbf{u}$, strain, or crack-aperture diagnostic; residual blocked until positions, zero date, sign convention, units, and uncertainty are supplied.
Laser scans	Registered niche-surface displacement or scan-difference fields.	Minutes state a 2026-04-20 statistical laser-scan update was sent; the provided catalogue has only surface geometry.	Surface-displacement diagnostic; blocked until registered products, masks, reference epochs, and registration uncertainty are supplied.
Levelling	Vertical displacement at survey points.	A small slide-summary set of height differences from the 2022-08-29/30 reference to the 2026-03 campaign.	Vertical $\mathbf{u}$ validation target; diagnostic only until full survey table, coordinates, reference frame, and covariance are supplied.
Auxiliary Geoscope / evapometer	RH, temperature, door/opening times, suction, evaporation or climate context.	Minutes and support geometry; no residual-ready auxiliary export here.	Boundary-condition provenance and disturbance filtering; not an interior OGS residual by default.

Time of measurement. The provided package contains qualitative 2026 statements, a source crackmeter trend plot, a source mini-piezometer trend plot, and one numeric levelling slide summary. The May 2026 minutes state that the 2026 Geoscope update covered RH, temperature, door opening times, extensometer, mini-piezometer, crackmeter, suction, and laser scans, and that horizontal/vertical extensometers BCD-A9/B10 in the closed niche failed after September 2025 [6, PDF p. 4]. The levelling slides report height differences from the first survey on 2022-08-29/30 to the 2026-03 campaign [7, PDF pp. 2, 5].

Measured value, units, and model link. Potential model ties are straightforward once data exist. Extensometers constrain axial displacement or strain from $\mathbf{u}$. Crackmeters constrain crack aperture or relative displacement. Laser scans constrain surface displacement or scan-difference fields. Miniprisma and convergence points constrain point or pair displacement. Levelling constrains vertical displacement. Evapometer or climate proxies should enter only through boundary forcing, not as interior residuals. For levelling, the current paper keeps the slide summary as text rather than as a figure: it reports height differences in millimetres between the 2022 reference survey and the 2026-03 campaign, mixes Galerie 18, closed-twin, and open-twin points, and lacks the full survey table. The relevant message for the open-niche note is therefore only that millimetre-scale vertical motion is present and could become a vertical displacement validation target after coordinates, reference frame, and covariance are supplied.

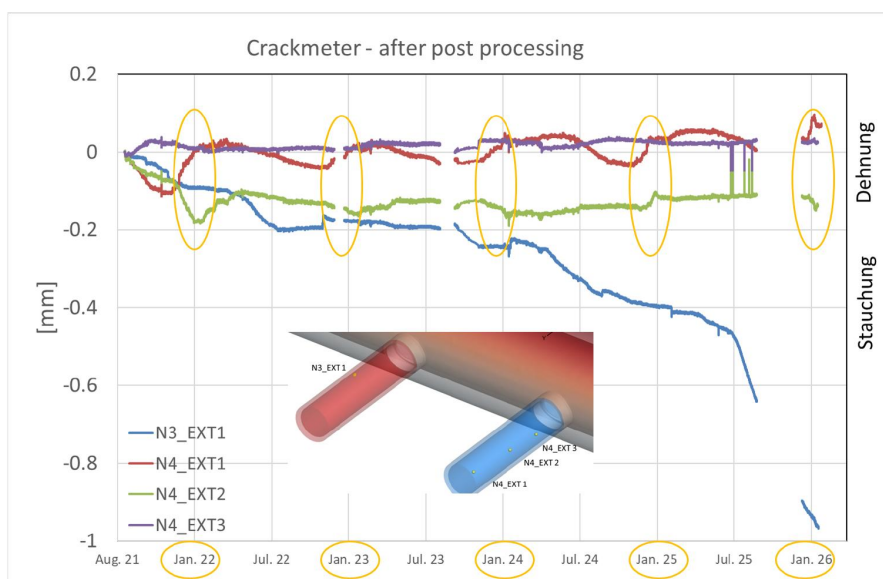


Figure 2.14: Crackmeter trend evidence from the 2026 technical discussion slides. The plotted post-processed deformation is in millimetres and spans roughly August 2021 to January 2026. The slide shows seasonal markers and an instrument inset. It is useful trend evidence, but not a residual-ready export because the paper still lacks positions, zero/reference convention, sign convention, uncertainty, and quality flags.

Uncertainty and reliability. The HM streams remain mostly blocked for hard residuals. The source scan found layout geometry, laser surface geometry, levelling slide rows, and report/minute evidence, but no residual-ready Geoscope time series, laser statistical product, or full levelling survey table [5, provided HM source files and generated trend figures]. A hard residual requires timestamps or survey epochs, values, units, support geometry, reference-zero conventions, sign conventions, calibration, quality flags, failure flags, and uncertainty or covariance. The levelling slide gives a detectable displacement of 0.1–0.2 mm at 95% confidence. Across the slide-summary points the largest settlement is CDA-O1 (−2.1 mm) and the largest uplift is the closed-twin point CDA-C4 (+1.3 mm). Because only the open-twin values belong to the current open-niche comparison and the slide still lacks the full survey table and reference-frame metadata, these numbers remain qualitative validation context rather than weighted residuals [7, PDF pp. 5–6].

Current computational use.

Crackmeter and comparable HM streams are not active likelihood terms in the present report. They should be retained as qualitative validation, structural priors, or future residual packages until numeric exports, reference zeros, instrument axes, sign conventions, supports, and uncertainties are supplied. With the available source data now, niche-surface cracks and other HM streams should mostly be validation gates, qualitative priors, or future residual packages. The crackmeter source plot can test whether a model candidate gives plausible opening or closure trends only after the measured component, zero date, sign convention, instrument axis, and support are known. Extensometers, convergence, miniprisma, levelling, and laser scans have clear OGS displacement ties, but the current workflow keeps mechanical parameters and prestress fixed. Therefore these streams should first reject mechanically implausible hydraulic candidates, support EDZ/fracture scenarios, or define quality/disturbance flags; they should not yet drive a coupled HM calibration [29, Table 1] [26, abstract and Sec. 2] [6, PDF p. 4 and p. 8] [5, provided HM source files and generated trend figures].

If numeric exports are supplied, the operators are standard but stream-specific: extensometers use projected displacement or strain along an instrument axis; convergence uses point-pair displacement; crackmeters use relative displacement or crack opening; levelling uses vertical displacement; laser scans use registered surface differences with masks and registration covariance. Auxiliary climate, door, temperature, evapometer, or Geoscope context should enter as boundary provenance, quality flags, or disturbance filters rather than interior residuals.

Grounding questions.

Table 2.10: Grounding questions for Geoscope, crackometer, and other HM monitoring.

Question. [Q080]	What does each HM stream measure?
Answer.	Extensometers measure axial deformation, crackmeters measure crack opening or relative displacement, levelling measures vertical displacement, laser scans measure surface differences, and auxiliary streams describe boundary or quality context; each source export must confirm the exact quantity.
Question. [Q081]	How should crackmeters link to OGS?
Answer.	Compare to relative displacement $(\mathbf{u}_a - \mathbf{u}_b) \cdot \mathbf{n}$, where $\mathbf{n}$ is the local crack-normal or instrument-axis direction, near-crack strain, or a crack-aperture diagnostic only after the component and sign convention are known.
Question. [Q082]	Can visible cracks justify permeability changes?
Answer.	Yes as an EDZ/fracture prior or scenario, but not as a weighted residual unless a measured aperture or displacement operator is supplied.
Question. [Q083]	How should levelling, extensometers, and laser scans be used now?
Answer.	As diagnostic HM validation or rejection checks for candidate permeability and boundary scenarios, because mechanical-parameter fitting is not currently grounded as a model component.
Question. [Q084]	Which other measurements can enter the OGS workflow?
Answer.	Any exported pressure, displacement, strain, crack-width, laser, climate, temperature, evapometer, geology, or support-geometry stream can enter as an input, residual, prior, mask, quality filter, or caveat once its support, time, units, and uncertainty are explicit.
Question. [Q085]	What would fully ground HM residuals?
Answer.	Machine-readable values with timestamps or survey epochs, coordinates/supports, units, reference-zero and sign conventions, instrument axes, failure/quality flags, uncertainty or covariance, and a 3D-to-2D comparison policy.

Chapter 3

Numerical Examples: ERT-Conditioned Material Fields

3.1 Purpose and Scope

This chapter records the numerical examples that have now been run for the CD-A Niche 4 ERT problem. It is intentionally an ERT chapter. The model equations, boundary conditions, and measurement inventory remain those of Chapters 1 and 2; the results here test how far the ERT support data can be matched by changing a small set of material-field and observation-model parameters.

Three messages are carried forward. First, the ERT comparison prefers a time shift of about three months relative to the OGS time origin, while the active open-niche seasonal boundary curve is not shifted in the model input. Second, a single global Archie law does not fit the ERT supports as well as support-wise Archie calibration. Third, the current successful ERT parameterisation uses a permeability-magnitude random field and a porosity random field. The tensor shape is still controlled by scalar angle and ratio parameters; the available results do not show that releasing a fully spatial tensor field is the missing step.

3.2 Parameterisation and Observation Model

The permeability field reported in the figures is K_{mag} , the first principal intrinsic-permeability value before tensor rotation. It is not the determinant, trace, or norm of the tensor. The random field is represented as

$$\log_{10} K_{\text{mag}}(\mathbf{x}) = k_0 + \sum_{j=1}^{100} \xi_j^K \psi_j(\mathbf{x}), \quad (3.1)$$

where k_0 is the global $\log_{10}$ -permeability level, ψ_j are spatial basis functions, and ξ_j^K are permeability-field coefficients. The OGS tensor input $\mathbf{K}$ is then reconstructed from K_{mag} , a sampled tensor rotation α , and a sampled principal-value ratio $q = K_2/K_1$. Before rotation, the principal values are

$$K_1(\mathbf{x}) = K_{\text{mag}}(\mathbf{x}), \quad K_2(\mathbf{x}) = q K_{\text{mag}}(\mathbf{x}).$$

With

$$R(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix},$$

the cell-wise tensor passed through the OGS `k_i_rd` field is

$$\mathbf{K}(\mathbf{x}) = R(\alpha) \begin{bmatrix} K_{\text{mag}}(\mathbf{x}) & 0 \\ 0 & q K_{\text{mag}}(\mathbf{x}) \end{bmatrix} R(\alpha)^T. \quad (3.2)$$

Equivalently, the tensor components written to OGS are

$$K_{11} = K_{\text{mag}}(\cos^2 \alpha + q \sin^2 \alpha), \quad (3.3)$$

$$K_{22} = K_{\text{mag}}(\sin^2 \alpha + q \cos^2 \alpha), \quad (3.4)$$

$$K_{12} = K_{21} = K_{\text{mag}}(1 - q) \sin \alpha \cos \alpha. \quad (3.5)$$

If the same tensor is written with geometric-mean magnitude $k_g(\mathbf{x})$ and anisotropy ratio $r = K_1/K_2$, then $k_g = K_{\text{mag}}\sqrt{q}$, $r = 1/q$, and $\mathbf{K} = R(\alpha) \text{diag}(k_g\sqrt{r}, k_g/\sqrt{r})R(\alpha)^T$.

The porosity field uses a bounded 100-mode random field around a sampled mean:

$$d_n(\mathbf{x}) = \sum_{\ell=1}^{100} \eta_\ell \varphi_\ell(\mathbf{x}), \quad (3.6)$$

$$n(\mathbf{x}) = n_{\min} + \frac{n_{\max} - n_{\min}}{1 + \exp[-\{\beta_n + 1.25(d_n(\mathbf{x}) - \bar{d}_n)\}]}, \quad (3.7)$$

where η_ℓ are porosity-field coefficients, φ_ℓ are the spatial basis functions, $\bar{d}_n$ is the mesh mean of d_n , $n_{\min} = 0.05$, $n_{\max} = 0.175$, and β_n is chosen so that the mesh mean equals the sampled mean porosity ϕ_0 .

The sampled material state is therefore not a fully free tensor field. For permeability, K_{mag} is spatially random through k_0 and ξ_j^K , while α and q are random scalar controls shared by the mesh. For porosity, ϕ_0 is a random scalar mean and η_ℓ are the spatial random-field coefficients. Table 3.1 also lists the fixed boundary controls and fitted observation-model coefficients.

Table 3.1: Prior coordinates and fixed controls used by the current ERT-conditioned examples.

Coordinate	Status and prior or treatment	Short role
k_0	sampled; $\mathcal{N}(-21.02, 5.1^2)$	Global level in Eq. (3.1).
$\xi_1^K, \dots, \xi_{100}^K$	sampled; independent $\mathcal{N}(0, 1)$	Permeability-magnitude random-field coefficients.
α	sampled; $\mathcal{N}(139.9, 5^2)$, bounded to 0–180°	Rotation in Eq. (3.2).
$\log_{10} q$	sampled; $\mathcal{N}(\log_{10} 0.45, 0.30^2)$, bounded by $\log_{10}[0.1, 0.9]$	Tensor ratio in Eq. (3.2).
ϕ_0	sampled; truncated $\mathcal{N}(0.1504, 0.012^2)$ on 0.05–0.175	Mean target for Eq. (3.7).
$\eta_1, \dots, \eta_{100}$	sampled; independent $\mathcal{N}(0, 1)$	Porosity random-field coefficients.
Δt_{ERT}	sampled; truncated $\mathcal{N}(96.082, 10^2)$ d on 66.082–106.082 d	ERT observation-time alignment.
$\Delta t_{\text{niche}}, s_{\text{niche}}, c_{\text{niche}}$	fixed to 0 d, 1, and 0 MPa in this run	Open-niche suction-curve shift, scale, and offset.
a_s, b_s	fitted per support, not sampled; $a_s \in [-0.7, 1.3]$, $b_s \in [-2.0, -0.2]$	Local Archie intercept and slope in Eq. (3.8).

For each ERT support s and time t_j , let $y_{s,j}$ be measured $\log_{10}$ -resistivity and $h_s(t_j; \vartheta)$ the support response from OGS or the trained surrogate for a parameter vector ϑ . Two Archie observation models are compared:

$$\hat{y}_{s,j} = a_s + b_s h_s(t_j; \vartheta) \quad \text{support-wise, or local, Archie,} \quad (3.8)$$

$$\hat{y}_{s,j} = a + b h_s(t_j; \vartheta) \quad \text{global Archie.} \quad (3.9)$$

The local model is flexible because every support has its own intercept and slope. The global model is the stricter test: one electrical calibration must explain all 20 support curves.

3.3 Time Alignment

Figure 3.1 shows the posterior distribution of the ERT time shift. The posterior median is 93.3 d and the 5–95% interval is 78.9–104.3 d. The best-likelihood retained sample is close to 96.1 d. This is not a material parameter: it says that the ERT observation time axis is displaced relative to the OGS time origin. Because the active open-niche boundary curve is fixed at zero shift in these examples, the ERT responses and the seasonal boundary forcing should not be read as already time-consistent.

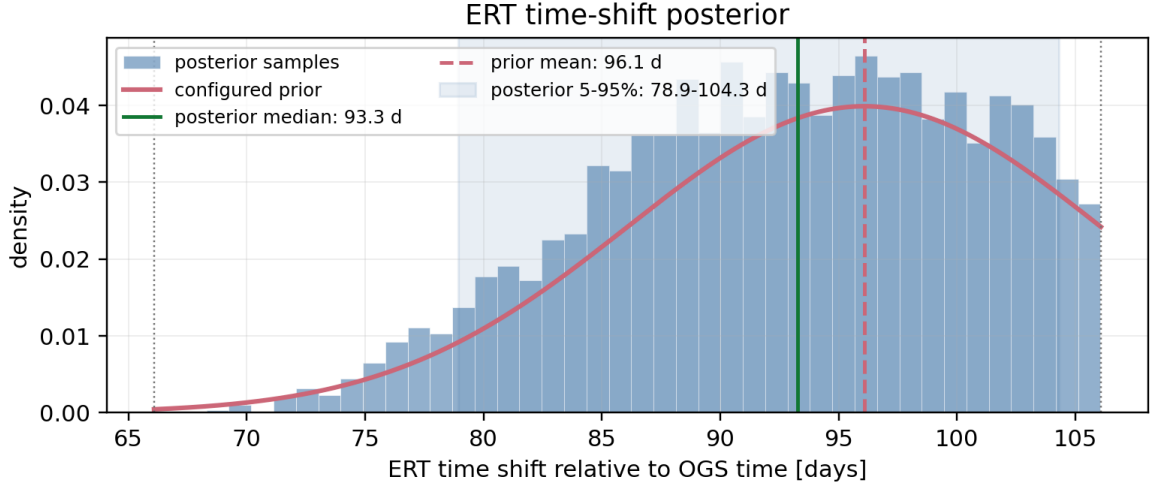


Figure 3.1: Posterior distribution of the ERT time-shift parameter. The configured prior is shown with the retained posterior samples; the posterior remains centred near a three-month shift.

3.4 Local and Global Archie

The strongest current ERT fits use local Archie calibration. In the retained raw- $\log_{10}$ posterior, the best-likelihood sample has MAE 0.0241 $\log_{10}$ -resistivity units and the posterior-median fit has mean absolute median residual 0.0248. The exact local-Archie ERT20 runs give the same scale of fit: about 0.024–0.028 MAE. By contrast, the global-Archie free- t_0 diagnostic did not become a posterior run; its best pre-crash candidate stays near 0.0825 MAE.

Observed, best-fit, and posterior-predictive responses

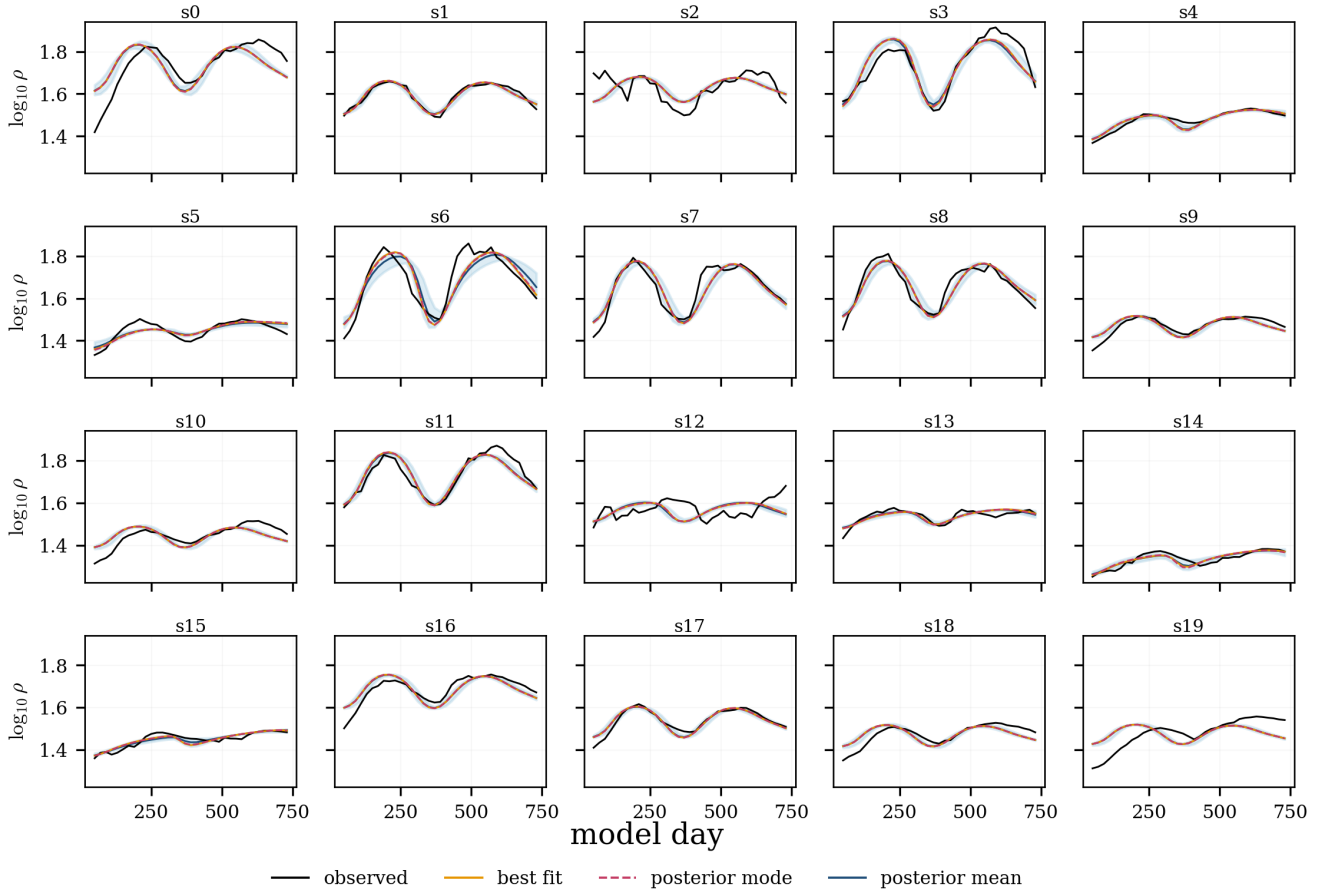


Figure 3.2: Local-Archie ERT20 fit. Each ERT support has its own fitted a_s, b_s pair, and the resulting support-wise fit reaches 0.028417 MAE in the exact exploratory run.

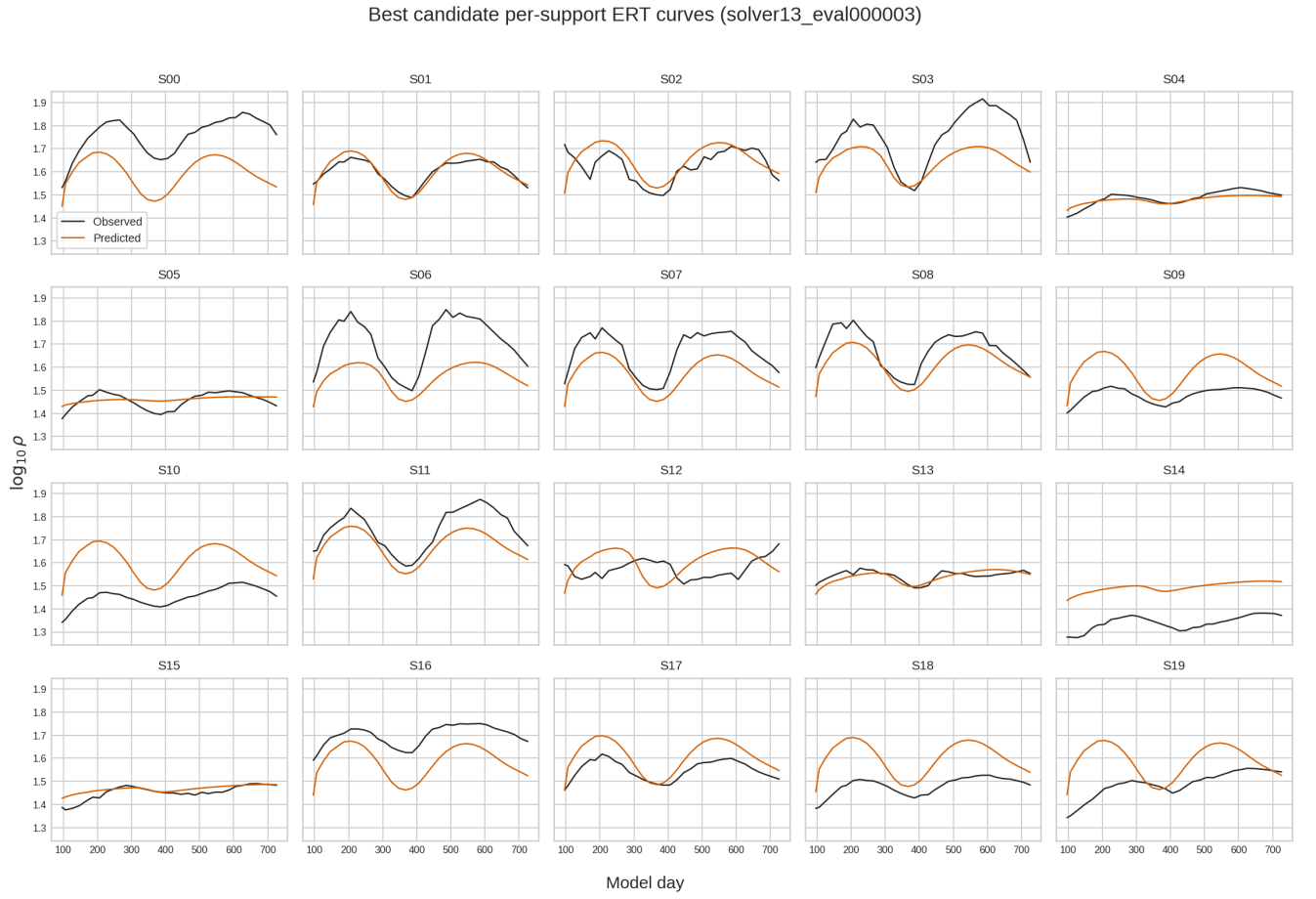


Figure 3.3: Global-Archie free- t_0 diagnostic. One a, b pair is used for all supports; the best pre-crash exact candidate has MAE about 0.0825.

The circular Archie summary in Fig. 3.4 explains why this matters. The fitted coefficients are not just noisy constants: the posterior medians and 5–95% bands vary by support angle and by the 15 cm versus 30 cm rings. This has two possible interpretations. It may be compensating for an imperfect ERT support operator, electrode/contact effects, shallow current paths, or electrical anisotropy not represented in the current observation model. It may also be a real indication that the hydraulic-to-electrical mapping is local near the niche. Either way, the local-Archie residual should not be interpreted as a pure permeability-field likelihood until the ERT support and electrical model are settled.

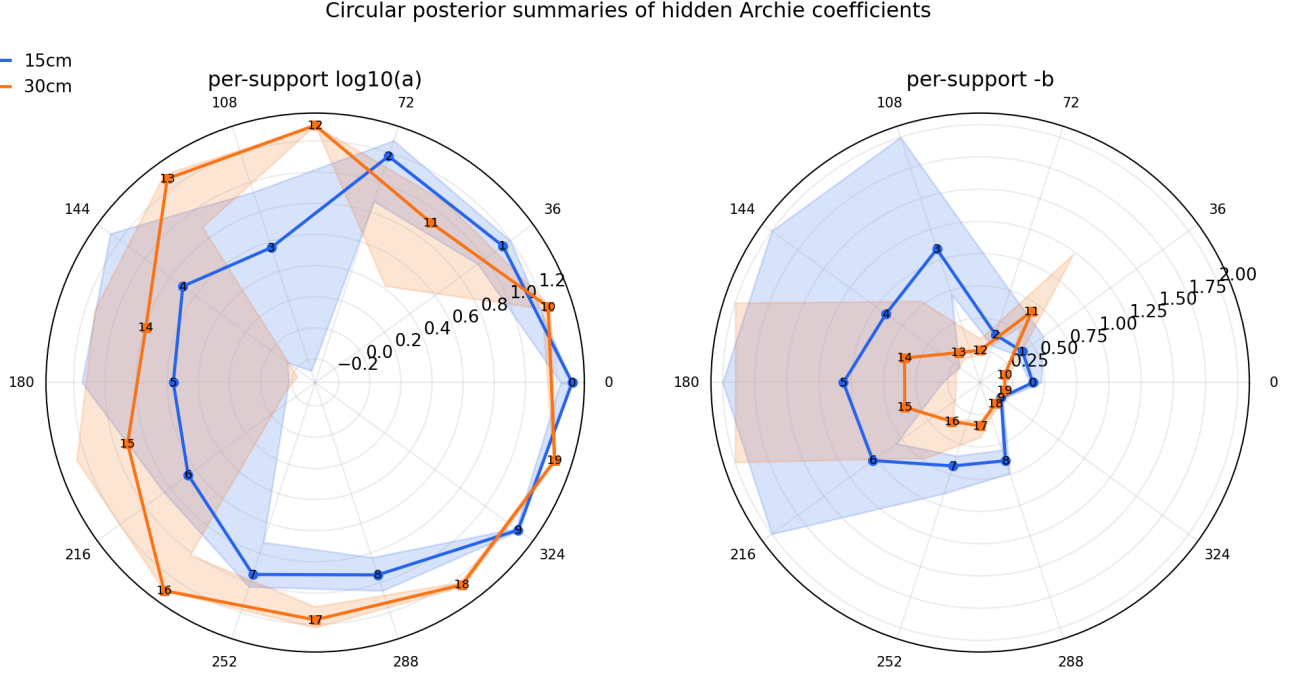


Figure 3.4: Circular posterior summaries of local Archie coefficients. Lines show posterior medians and shaded bands show 5–95% intervals for the two ERT support rings. Spatial structure in these nuisance coefficients is a measurement-model result, not a direct material-field result.

3.5 Permeability and Porosity Fields

The field result should be read as an ERT-conditioned material-field diagnostic. Earlier permeability-magnitude examples showed that a spatial $\log_{10} K_{\text{mag}}$ field improves the ERT response, but the best current raw- $\log_{10}$ fit uses both the permeability-magnitude field and the bounded porosity field in Eqs. (3.1) and (3.7). The parameterisation is therefore best described as the minimal successful one seen in these files: scalar tensor orientation and ratio, plus two random fields, K_{mag} and n . The available results do not justify promoting a fully spatial tensor-random-field model as better supported than this smaller parameterisation.

In the retained posterior-field summary, the posterior mean $\log_{10} K_{\text{mag}}$ field has spatial median -16.88 , 5–95% spatial range -21.66 to -10.14 , and posterior standard-deviation median 2.31 . These values show that ERT is pushing a broad permeability-magnitude field, not a narrowly identified final map.

The porosity posterior is also spatially structured. The posterior mean porosity field has spatial median 0.155 , 5–95% spatial range 0.119 – 0.172 , and posterior standard-deviation median 0.0147 . This is not the fixed `n_rd` porosity field from the base XML; it is an ERT-conditioned random-field result and still needs exact-OGS checks and independent water-content or porosity support before it becomes a released material field.

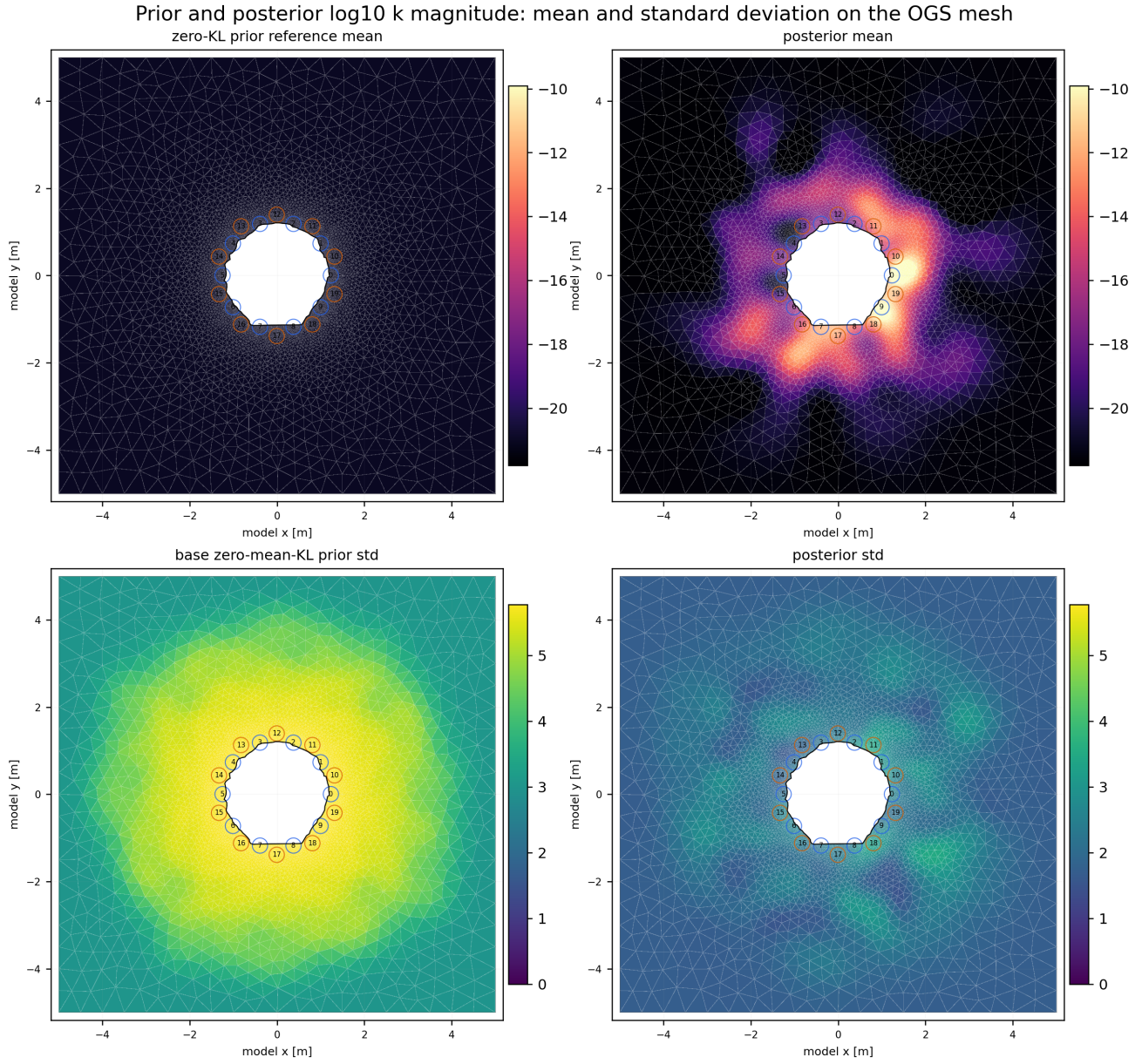


Figure 3.5: ERT-conditioned log₁₀ K_{mag} field. The figure compares the zero-KL prior reference and posterior summaries for permeability magnitude on the OGS mesh.

Prior and posterior porosity: mean and standard deviation on the OGS mesh

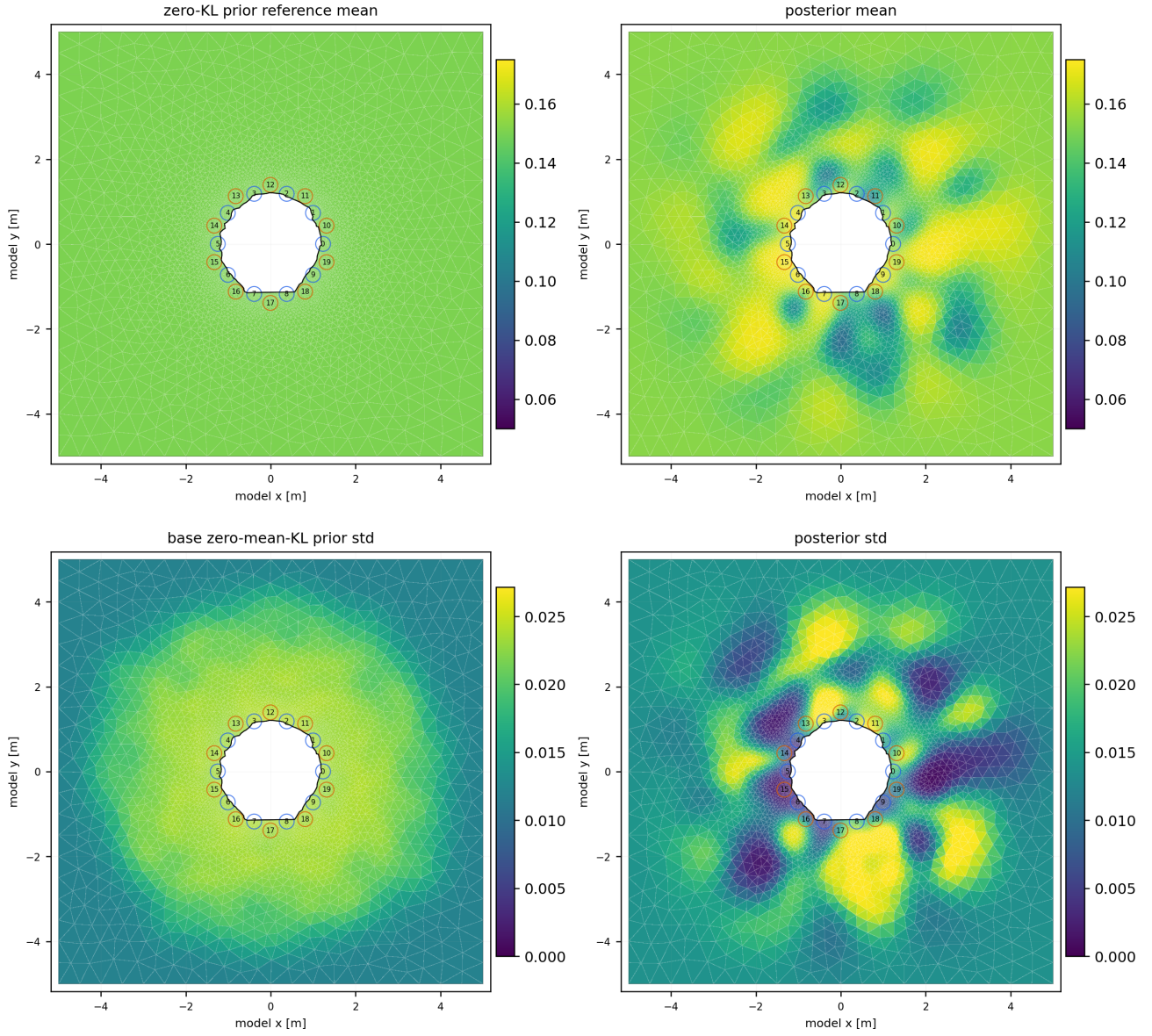


Figure 3.6: ERT-conditioned porosity field. The panels show the zero-KL prior-reference mean, posterior mean, base zero-mean-KL prior standard deviation, and posterior standard deviation on the native OGS mesh.

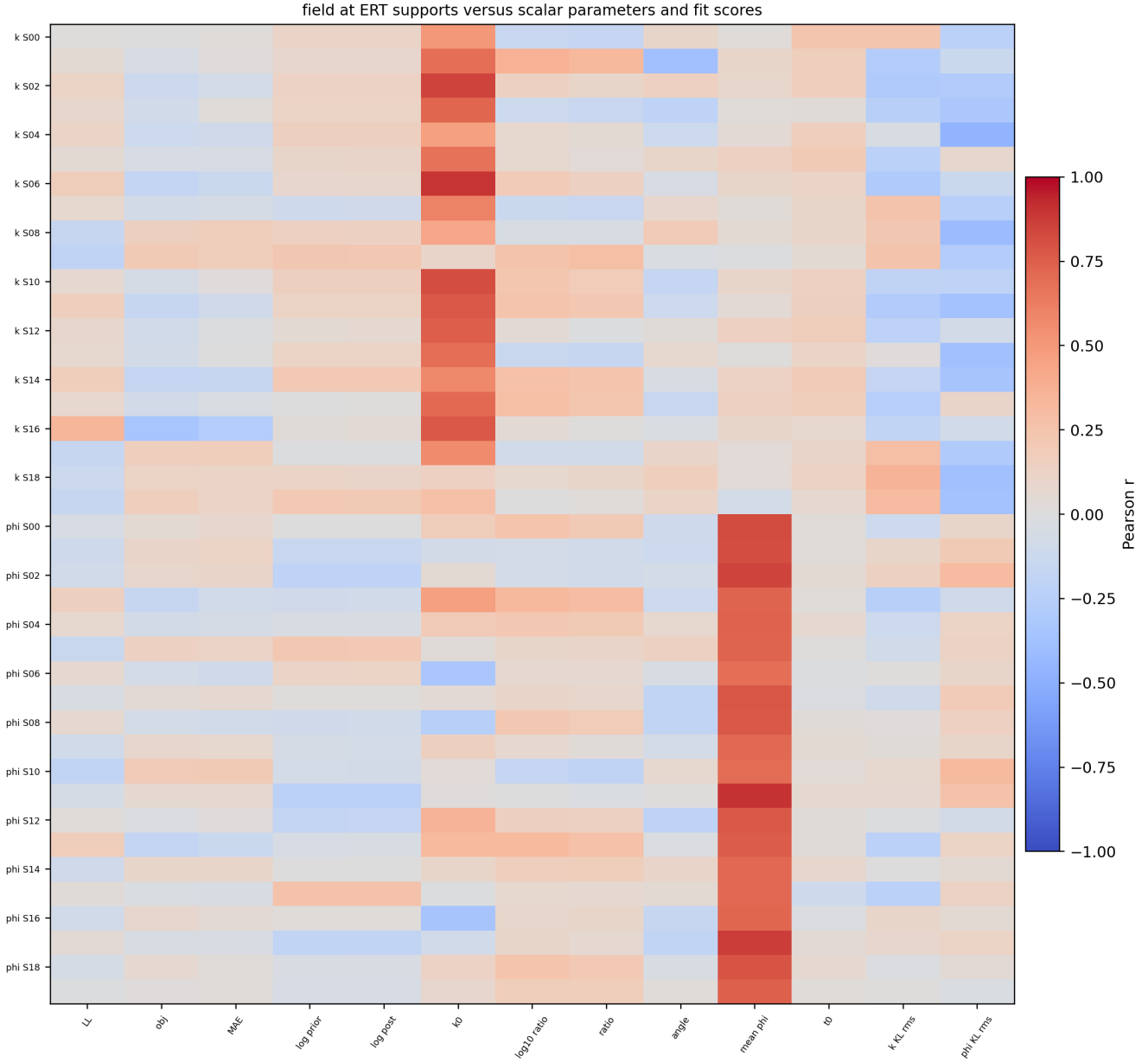


Figure 3.7: Posterior correlations between ERT-support random-field values and scalar parameters or fit scores. The permeability-support variables correlate most strongly with k_0 ; porosity-support variables correlate most strongly with the mean porosity control.

3.6 Implications for the Rest of the Report

The numerical examples move ERT from a future stream to a demonstrated calibration target, but only within a narrow observation model: 20 ERT supports, 33 times, an explicit time-alignment parameter, and Archie calibration. They do not assimilate NMR, Taupe/TDR, RH/suction, direct pressure, fractures, crackmeters, levelling, or other hydro-mechanical streams.

The practical interpretation is therefore conservative. Keep K_{mag} as the common permeability field written through `k_i_rd`; use the porosity random field as an ERT-conditioned diagnostic until it is checked by exact OGS and non-ERT evidence; and treat the local/global Archie contrast as the main current ERT gate. The next robust comparison should fix the support mask, time-axis convention, covariance/down-weighting, and electrical anisotropy treatment before turning these ERT-conditioned fields into a final all-measurement material map.

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